

RESEARCH ARTICLE

Performance evaluation of inerter-based dampers for vortex-induced vibration control of long-span bridges: A comparative study

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Funding information

China Postdoctoral Science Foundation, Grant/Award Number: 2018 T110022; National Natural Science Foundation of China, Grant/Award Numbers: 51708011, 51778023

Summary

Inerter-based dampers (IBDs) have been proposed to enhance the efficiency of conventional tuned mass dampers (TMDs) in controlling wind-induced vibration of civil structures recently. In this paper, four different IBDs, with each featuring a specific combination of dashpot-spring elements in series or in parallel with the inerter device, are applied to control the vortex-induced vibration (VIV) of long-span bridges. The governing equations of the bridge-IBD system under the effect of VIV are established, and the strategy to optimize the IBD parameters is proposed. The performances of different IBDs in terms of mitigating the VIV response of the deck, the stroke of mass block, the static deformation of spring due to gravity, the reaction force of IBDs on the bridge deck, and the robustness of the system (the control efficiency against the deviation of design parameters from their optimal values) are systematically evaluated and compared. Through this comparative study, the superiorities of IBDs to conventional TMDs in bridge VIV control are illustrated, and two feasible layouts of IBDs that are suitable for bridge VIV control are suggested. On the basis of these results, the guideline for VIV-resistance design of long-span bridges with the help of IBDs is also proposed.

KEYWORDS

comparative study, inerter-based damper, long-span bridges, tuned mass damper, VIV control

1 | INTRODUCTION

Vortex-induced vibration (VIV) is a typical wind-induced hazard for long-span bridges, which often occurs at low and moderate wind velocities.¹⁻³ The undesirably large amplitude of VIV seriously threatens the safety of bridge users and decreases the fatigue lifetime of the bridge. In extreme circumstances, it may cause catastrophic damages to important structural elements or even collapse of the whole structure. The VIV-resistance design of long-span bridges is thus a major concern for bridge engineers.

To control the adverse VIV of bridges, many control schemes have been proposed. Among these, the tuned mass damper (TMD) has been widely adopted, for example, in the Trans-Tokyo Bay Bridge,² the Rio-Niteroi Bridge,⁴ the Kessock Cable-Stayed Bridge,⁵ the Volgograd Bridge,⁶ and the Jindo Bridge.⁷ The main advantage of this scheme is that

it can be installed inside the main deck and will not influence the esthetics of the bridge. Thus, it is often adopted when the application of aerodynamic control schemes is restricted, or as a supplemental plan to aerodynamic control schemes to further enhance the control efficiency.

However, the application of TMD in bridge VIV control faces some practical challenges. The first challenge is the large static deformation of TMD spring due to gravity. The static deformation of the spring can be estimated by $x_0 = g/\omega_{TMD}^2$, where g is the acceleration of gravity and ω_{TMD} is the circular frequency of the TMD.⁸ It is well-known that the frequency of TMD is generally tuned to the frequency of the primary structure to resonate with it. For long-span bridges, the natural frequencies of the first several modes are quite low; the static deformation of TMD spring thus can be quite large, which makes it difficult to install the TMDs inside the bridge deck. The second challenge is the dimension of the TMD mass block. The effectiveness of TMD highly depends on the percentage of the auxiliary mass, and a large percentage of TMD mass generally results in better control performance. However, increasing TMD mass increases not only the cost but also the dimension of the mass block, which is again often restricted by the internal space of the deck. The third challenge is the possible large stroke of the TMD mass block. In the TMD-based vibration control method, the vibration energy of the primary structure (i.e., the bridge) is transferred to the TMD mass block, which makes the stroke of the mass block much larger than the vibration amplitude of the primary structure. A large stroke of the mass block may result in pounding problem once the clearance between the attached mass and the primary structure is not enough. All these issues require that the internal space of the deck should be large enough. However, for long-span bridges, the streamlined box decks are normally adopted, and the deck height and internal space of these streamlined box decks are generally limited, which makes it difficult to implement the TMDs.

Recently, a two-terminal mechanical device dubbed inerter was introduced to the vibration control of structures.⁹⁻¹² This device is able to generate a resisting force proportional to the relative acceleration between its two terminals, and this force can be several orders of magnitude higher than the one generated by its physical mass.^{13,14} Later on, this device was combined with conventional vibration absorbers to further improve their performances. Ikago et al.¹⁵ combined this device with a viscous damper and proposed the viscous mass damper and the tuned viscous mass damper systems. Lazar et al.¹⁶ incorporated this device with damping and stiffness elements and proposed the tuned inerter damper system. Marian and Giaralis¹⁷ combined this device with conventional TMD and proposed the TMD inerter (TMDI) system. Because there exist different combinations of inerter with the spring and damper elements (refer to Figure 1), Wen et al.,¹⁸ Pietrosanti et al.,¹⁹ and Hu and Chen²⁰

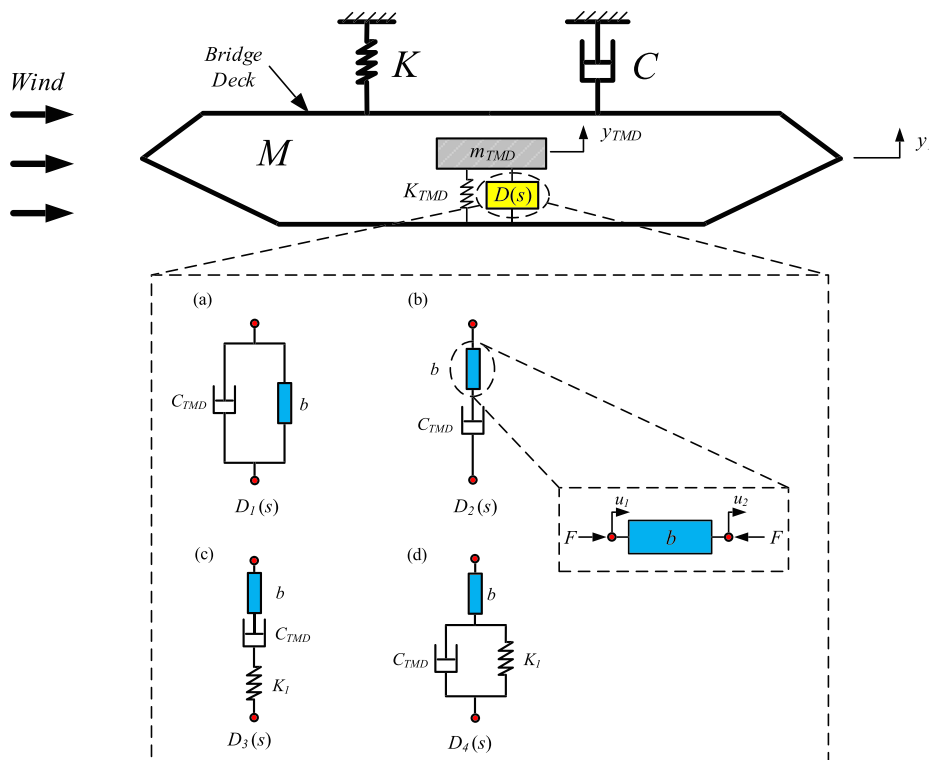


FIGURE 1 Schematic diagram of the bridge-IBD system. IBD, inerter-based damper

compared the performances of different combinations of inerter-based dampers (IBDs). All these studies demonstrated the efficiency of IBDs in reducing seismic-induced vibration of civil structures. Very recently, these dampers were also applied to the vibration control of cables,^{21–23} onshore²⁴ and offshore²⁵ wind turbines, and offshore platforms.^{26,27}

Using IBD to control wind-induced vibration of engineering structures is relatively rare. To the best knowledge of the authors, the first attempt to apply IBD in wind-excited vibration control of civil structures was conducted by Giaralis and Petrini,²⁸ in which the TMDI system was adopted to control the crosswind induced vibration of tall buildings. It was found that TMDI is more effective than TMD, and the inclusion of the inerter can dramatically reduce the TMD stroke. Very recently, Xu et al.²⁹ applied TMDI to control VIV of long-span bridges. The control performance and robustness (the control efficiency against the deviation of design parameters from their optimal values) of the TMDI system were investigated and compared with that of the conventional TMD system. It was found that the TMDI system can significantly reduce the static stretching of the spring due to gravity as well as the oscillation amplitude of the mass block.

As discussed above, the effectiveness of one combination of IBD, namely TMDI, was adopted to control the VIV of long-span bridges in Xu et al.²⁹ In fact, the inerter, spring, and dashpot can have different combinations as discussed, and each combination may have different effectiveness. The aim of the present study is to investigate the effectiveness of different IBDs to control the VIV of long-span bridges and identify the most effective and practical way in the design. To this end, four different IBDs are investigated. The governing equations of each bridge–IBD system are established, and the optimized parameters of each IBD are obtained. The performances of different IBDs in terms of mitigating the VIV response of the deck, the stroke of mass block, the static deformation of spring due to gravity, the reaction forces of IBDs on the bridge deck, and the robustness of the system are systematically evaluated and compared. The organization of this paper is as follows: In Section 2, the governing equations for each bridge–IBD system are established, and the optimum design strategy is proposed. The control performances of different IBDs are systematically evaluated and compared through a numerical case study in Section 3. In Section 4, the performances of IBDs in bridge VIV control at a variety of mass ratios and inertance ratios are further discussed. In Section 5, the guidance for VIV-resistance design of long-span bridges with the help of IBDs is established. Finally, some concluding remarks are drawn in Section 6.

2 | BRIDGE VIV CONTROL WITH IBDS

2.1 | Governing equations of the bridge–IBD system

Figure 1 shows the bridge–IBD systems investigated in the present study. The outline border denotes the bridge main deck with M denoting the mass of the deck and K and C representing the stiffness and damping. This bridge will vibrate in the vertical direction under the horizontal wind due to the vortex-shedding effect. To control VIV, four different IBD arrangements are considered. In particular, m_{TMD} and K_{TMD} are the auxiliary mass and stiffness similar to the conventional TMD system, and the damping element in the conventional TMD system is replaced by the network D_1 , D_2 , D_3 , and D_4 , which include different combinations of the stiffness element (K_1), the damping element (C_{TMD}), and the inerter (b) as shown.

In the present study, four different arrangements denoting IBD1 to IBD4 are considered on the basis of the possible combinations of the damper, spring, and inerter as shown in Figure 1. In particular, the arrangement IBD1 (D_1 plus m_{TMD} and K_{TMD}) is identical with the TMDI system proposed by Marian and Giaralis,¹⁷ and the other configurations are based on the inerter mechanical networks proposed in Hu and Chen.²⁰

The resisting force of the inerter device can be calculated by⁹

$$F = b(\ddot{u}_1 - \ddot{u}_2), \quad (1)$$

where $\ddot{u}_1 - \ddot{u}_2$ is the relative acceleration between the two terminals of this device; b is dubbed inertance (apparent mass). The inerter device can be practically realized in several ways, for example, via the rack-and-pinion,³⁰ ballscrew,³¹ or hydraulic elements.³²

Taking the ballscrew type of inerter as an illustration (as shown in Figure 2), the inertance of this system can be expressed as

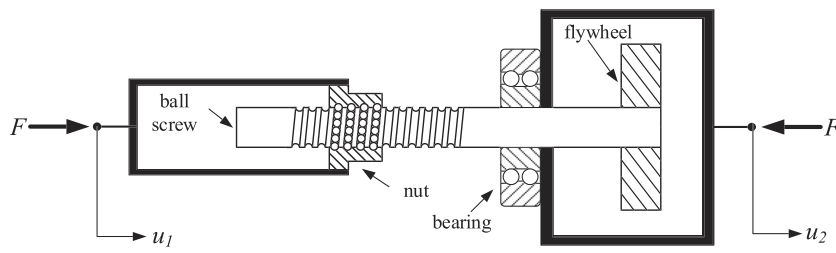


FIGURE 2 Schematic diagram of the ballscrew type inerter

$$b = \left(\frac{2\pi}{P} \right)^2 J_{\text{total}}, \quad (2)$$

where P is pitch size of the screw; J_{total} is the total mass moments of inertia of the screw rod and flywheel along the central axis of the screw. The lateral movement of the ballscrew generates rotational movement in the screw rod and flywheel, and the rotational inertia of this device can be adjusted through increasing (or decreasing) the diameter and thickness of the flywheel.

Two prototype ballscrew inerters manufactured by Smith et al.³³ as well as by the authors are illustrated by Figure 3. For an ideal inerter, the apparent mass can be several orders of magnitude higher than its physical mass.¹³ This is the key feature that makes this device extremely appealing for vibration control.

The equations of motion of the bridge-IBD system in the Laplace domain can be expressed as

$$\begin{cases} Ms^2 y_1 = F_{\text{wind}} + F_{\text{IBD}} - Ky_1 - Cs y_1, \\ m_{\text{TMD}} s^2 y_{\text{TMD}} = -F_{\text{IBD}}, \end{cases} \quad (3)$$

where s is the Laplace operator, m_{TMD} and M are the mass of the mass block and primary structure, respectively; C and K are the damping and stiffness of the bridge as discussed; y_1 and y_{TMD} are the displacement of the deck and mass block; F_{wind} is the VIV force acting on the bridge deck, and F_{IBD} is the force of the IBD system imposed on the primary structure, which can be calculated as follows:

$$F_{\text{IBD}} = [K_{\text{TMD}} + sZ(s)] \cdot (y_{\text{TMD}} - y_1), \quad (4)$$

in which $Z(s)$ is the impedance of the inerter-based mechanical networks,²⁰ which are summarized in Table 1.



(a) prototype ballscrew inerter manufactured by Smith et al.³³



(b) prototype ballscrew inerter manufactured by the authors

FIGURE 3 Prototype ballscrew inerters

TABLE 1 Impedance of each inerter-based mechanical network in Figure 1

IBD1	IBD2	IBD3	IBD4
$Z_1(s) = b \cdot s + C_{\text{TMD}}$	$Z_2(s) = \frac{1}{b s + C_{\text{TMD}}}$	$Z_3(s) = \frac{1}{b s + C_{\text{TMD}} + \frac{s}{k_1}}$	$Z_4(s) = \frac{1}{\frac{k_1}{s} + C_{\text{TMD}} + b s}$

Abbreviation: IBD, inerter-based damper.

By defining $y_2 = y_{\text{TMD}} - y_1$ as the relative displacement between the mass block and main deck, the above equations of motion can be rewritten as

$$s^2 \cdot \begin{bmatrix} M + m_{\text{TMD}} & m_{\text{TMD}} \\ m_{\text{TMD}} & m_{\text{TMD}} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + s \cdot \begin{bmatrix} C \\ Z(s) \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + \begin{bmatrix} K \\ K_{\text{TMD}} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} F_{\text{wind}} \\ 0 \end{bmatrix}. \quad (5)$$

According to field measurements, VIV is a single-mode vibration phenomenon and mainly occurs at the vertical modes of the bridge.¹⁻³ The VIV analysis of a bridge structure thus is normally conducted in the modal coordinate. To control the i th (i is a positive integer) mode VIV, the IBDs can be installed at the position of the generalized coordinate of that mode. Replacing s with $j\omega$ (where $j = \sqrt{-1}$ is the imaginary number) in Equation (5) and replacing the structural parameters by its modal properties, the equations of motion for the i th mode VIV control of the bridge can be expressed as

$$(j\omega)^2 \cdot \begin{bmatrix} M_i + \mu M_i & \mu M_i \\ \mu M_i & \mu M_i \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} + (j\omega) \cdot \begin{bmatrix} 2M_i \omega_{ni} (\xi_{ni} - \xi_{ae}) \\ Z(\omega) \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} + \begin{bmatrix} M_i \omega_{ni}^2 \\ K_{\text{TMD}} \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = \begin{bmatrix} F_i \\ 0 \end{bmatrix}, \quad (6)$$

where $M_i = \bar{m} \int_0^L \phi_i(x)^2 dx$ is the i th modal mass of the bridge, with $\bar{m}$ representing the equivalent mass of the deck per unit length, L the total length of the bridge, and $\phi_i(x)$ the modal shape of the deck; η_1 is the generalized coordinate of the bridge deck; η_2 is relative displacement of the IBD mass block; ξ_{ni} and ω_{ni} are the damping and frequency of the i th mode, respectively (with which, K and C in Figure 1 can be calculated); μ is the mass ratio of the mass block to the modal mass of the bridge, namely

$$\mu = m_{\text{TMD}} / M_i. \quad (7)$$

ξ_{ae} is the aeroelastic damping, which can be modeled by³⁴⁻³⁶

$$\xi_{ae} = K_a \frac{\rho D^2}{\bar{m}} \left[1 - \left(\frac{\sigma_{\eta_1}}{D a_L} \right)^2 \right], \quad (8)$$

where σ_{η_1} is the root mean square (RMS) of the deck's response η_1 , D is deck height, ρ is air density, and K_a and a_L are linear and nonlinear aeroelastic damping coefficients to be determined through experiment.

F_i in Equation (6) is the generalized lift force resulting from the vortex-shedding effect (forcing term in the VIV force that is independent with structural motion), the power spectral density (PSD) of which can be expressed as³⁶

$$S_{F_i}(\omega) = \frac{2q_v^2 D^3 \sigma_{CL}^2 \lambda}{\sqrt{\pi} \omega_s B} \exp \left[- \left(\frac{1 - \omega / \omega_s}{B} \right)^2 \right] \int_0^L \phi_i^2(x) dx, \quad (9)$$

where $q_v = \rho U^2 / 2$ is wind pressure with U denoting wind velocity, σ_{CL} is the RMS of the fluctuating lift coefficient due to vortex shedding, $\omega_s = 2\pi U S_t / D$ is vortex-shedding frequency, S_t is Strouhal number, B is load spectrum bandwidth, and λ is a dimensionless parameter that represents the coherency length-scale of vortices.

2.2 | Response of the bridge-IBD system

The frequency response functions of the bridge deck $H_{\eta_1}(\omega)$ and relative displacement of the IBD mass block $H_{\eta_2}(\omega)$ can be obtained through

$$\begin{bmatrix} H_{\eta_1}(\omega) \\ H_{\eta_2}(\omega) \end{bmatrix} = [R]^{-1} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad (10)$$

where

$$[R] = \begin{bmatrix} -\omega^2 \cdot (1 + \mu)M_i + j\omega \cdot 2M_i\omega_{ni} \cdot (\xi_{ni} - \xi_{ae}) + M_i\omega_{ni}^2 & -\omega^2 \cdot \mu M_i \\ -\omega^2 \cdot \mu M_i & -\omega^2 \cdot \mu M_i + j\omega \cdot Z(\omega) + K_{TMD} \end{bmatrix}. \quad (11)$$

The PSD of the deck ($S_{\eta_1}(\omega)$) and relative displacement of the IBD mass block ($S_{\eta_2}(\omega)$) then can be obtained:

$$\begin{bmatrix} S_{\eta_1}(\omega) \\ S_{\eta_2}(\omega) \end{bmatrix} = \begin{bmatrix} H_{\eta_1}^*(\omega) \cdot H_{\eta_1}(\omega) \\ H_{\eta_2}^*(\omega) \cdot H_{\eta_2}(\omega) \end{bmatrix} \cdot S_{F_i}(\omega), \quad (12)$$

where the symbol * denotes complex conjugation. The mean square values of η_1 and η_2 are therefore

$$\begin{bmatrix} \sigma_{\eta_1}^2 \\ \sigma_{\eta_2}^2 \end{bmatrix} = \int_{\omega} \begin{bmatrix} S_{\eta_1}(\omega) \\ S_{\eta_2}(\omega) \end{bmatrix} d\omega. \quad (13)$$

It should be noted that the aeroelastic damping ξ_{ae} in Equation (8) is a function of the RMS of the deck's response. Thus, the values of σ_{η_1} and σ_{η_2} should be calculated through iteration procedure. The main steps in the iteration are as follows:

1. First, an initial value of $\sigma_{\eta_{1,0}}$ is assumed (a value of $0.1D$ is adopted in this study);
2. Then, the value of aeroelastic damping ξ_{ae} is calculated by Equation (8), and the RMS of deck response $\sigma_{\eta_{1,1}}$ is calculated through Equations (10)–(13); and
3. If the difference between $\sigma_{\eta_{1,0}}$ and $\sigma_{\eta_{1,1}}$ is less than the tolerance ($1e-6$ in the present study), the iteration terminates; otherwise, the value of $\sigma_{\eta_{1,1}}$ is assigned to $\sigma_{\eta_{1,0}}$, and the above procedure is repeated.

It also should be noted that VIV is a self-excited phenomenon.³⁷ The stability of the bridge–IBD system under self-excited effect is determined by the sign of its two modal damping ratios. If the modal damping of the bridge–IBD system is negative, this system will be unstable.^{38,39} To avoid this problem, a constraint condition is added during the process of iteration. The constraint condition is that the calculated real parts of the complex eigenvalues of the bridge–IBD system should be negative.^{38,39} The eigenvalues of the bridge–IBD system are obtained through solving the following characteristic equation:

$$\det \begin{bmatrix} \lambda^2 \cdot (1 + \mu)M_i + \lambda \cdot 2M_i\omega_{ni} \cdot (\xi_{ni} - \xi_{ae}) + M_i\omega_{ni}^2 & \lambda^2 \cdot \mu M_i \\ \lambda^2 \cdot \mu M_i & \lambda^2 \cdot \mu M_i + \lambda \cdot Z(\lambda) + K_{TMD} \end{bmatrix} = 0. \quad (14)$$

The characteristic equations of each IBD system as shown in Figure 1 are summarized in Appendix A. The constrained iteration in the above procedure is conducted by using the *fmincon* command in the “Optimization Toolbox” in MATLAB.

2.3 | Optimal design and tuning procedure

As shown in Figure 1, the parameters defining the IBD performances are μ , b , K_{TMD} , C_{TMD} , and K_1 . For easy illustration, the following nondimensional parameters are introduced:

$$\left. \begin{aligned} \delta &= b/m_{\text{TMD}} = \beta/\mu : \text{inertance-to-mass ratio} \\ \gamma &= \omega_{\text{TMD}}/\omega_{ni} : \text{natural frequency ratio} \\ \nu &= \omega_b/\omega_{\text{TMD}} : \text{corner frequency ratio} \\ \xi_{\text{TMD}} &= \frac{C_{\text{TMD}}}{2\mu M_i \omega_{\text{TMD}}} : \text{damping ratio of the IBD} \end{aligned} \right\}, \quad (15)$$

where

$$\left. \begin{aligned} \beta &= b/M_i : \text{inertance ratio} \\ \omega_{\text{TMD}} &= \sqrt{\frac{K_{\text{TMD}}}{\mu M_i}} : \text{natural frequency of the IBD} \\ \omega_b &= \sqrt{\frac{K_1}{b}} : \text{corner frequency of the IBD} \end{aligned} \right\}. \quad (16)$$

The optimization of the IBD parameters contains two steps: (a) First, the proper mass ratio μ and inertance-to-mass ratio δ are determined in advance; (b) Second, the values of IBD design parameters γ , ν , and ξ_{TMD} are calculated through the optimum design strategy.

The control performance of each IBD can be evaluated by using the H_2 performance index I_{IBD} as proposed by Hu and Chen,²⁰ and it has the following form:

$$I_{\text{IBD}} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H_{\text{IBD}}(j\omega)|^2 d\omega, \quad (17)$$

where H_{IBD} is the dimensionless frequency response function of the main structure, and the expressions for each IBD system are summarized in Appendix B.

The optimum design of IBD is to find a set of parameters that generate the minimum value of I_{IBD} . During the optimization, a constrained condition is added on the basis of the fact that the values of γ , ν , and ξ_{TMD} should not be negative. The constrained minimization and optimization are conducted by using the genetic algorithm in MATLAB. In this study, a population size of 3,000 and a generation of 100 in the genetic algorithm are adopted on the basis of consideration of precision and efficiency.

3 | CASE STUDY AND EVALUATION

3.1 | Bridge model

To evaluate the control performances of different IBDs, a bridge investigated by Strommen and Hansen³⁶ is adopted as a case study. This bridge is a suspension bridge with a main span of $L = 595$ m. The superstructure is a streamlined box deck with a height of 2.5 m and width of 13.6 m. This bridge experienced VIV in its fourth vertical mode with a maximum VIV amplitude of 0.25 m. The modal damping and frequency are $\xi_{ni} = 0.24\%$ and $\omega_{ni} = 2.462 \text{ rad} \cdot \text{s}^{-1}$, respectively. The modal shape is $\phi_i(x) = \sin(4\pi x/L)$. The equivalent modal mass of the deck is $\bar{m} = 7460 \text{ kg/m}$.³⁶

As to the aeroelastic parameters, the bandwidth parameter B in Equation (9) is 0.2; the reduced lift coefficient $\sigma_{CL} \cdot (\lambda/B)^{0.5}$ is 3.92, and the nonlinear aeroelastic damping term a_L is 0.233. The linear aeroelastic damping term is calculated by³⁶

$$K_a = \frac{2.187}{(U/U_{cr} - 0.25)^2} \cdot \exp \left[\frac{-1}{(U/U_{cr} + 0.02)^{24}} \right] - 0.18, \quad (18)$$

in which $U_{cr} = D\omega_{ni}/(2\pi S_t)$ is the critical wind velocity of VIV; The Strouhal number of the deck is $S_t = 0.16$.

The normalized VIV responses of the bridge deck with respect to deck height are calculated, and they are shown in Figure 4. It can be seen that the maximum VIV response occurs at around $U/D\omega_{ni} = 1.06$. Thus, the aeroelastic parameters on this wind velocity are adopted for the IBD parameter optimization. Figure 5 shows the PSD of the main deck displacement at the most unfavorable wind velocity within the lock-in region ($U/D\omega_{ni} = 1.06$ as shown in Figure 4). It can be seen that the PSD function shows an obvious peak, and it corresponds to the natural frequency of the primary structure ($\omega_{ni} = 2.462 \text{ rad} \cdot \text{s}^{-1}$). This indicates that the vortex-shedding frequency (the external forcing frequency) at this wind velocity no longer follows the Strouhal law but is “locked-in” by the structural natural frequency. This phenomenon is a key feature of VIV, which validates that the adopted VIV model can well simulate the VIV property of the bridge.

3.2 | Control efficiencies of different IBDs

To evaluate the control efficiencies of different IBDs, two sets of mass parameters are selected: $\mu = 0.005$; $\delta = 2.5$ and $\mu = 0.05$; $\delta = 2.5$. The selection of mass ratio μ is based on the commonly adopted engineering practice; namely, for wind-induced vibration control, the mass ratio of the TMD system is normally less than 5% of the primary structure. The selected values (i.e., 0.5% and 5%) can cover the possible mass ratios used for bridge VIV control. As for the inertance-to-mass ratio δ , it is motivated by the fact that the commonly available inerter device can amplify the physical mass by several hundred times.³³ For example, if a mass amplification factor of 100 (i.e., inertance = 100) is selected, the required physical mass of the inerter is only 2.5% of the TMD mass when $\delta = 2.5$ is selected. However, as will be demonstrated in the following sections, with such a small mass

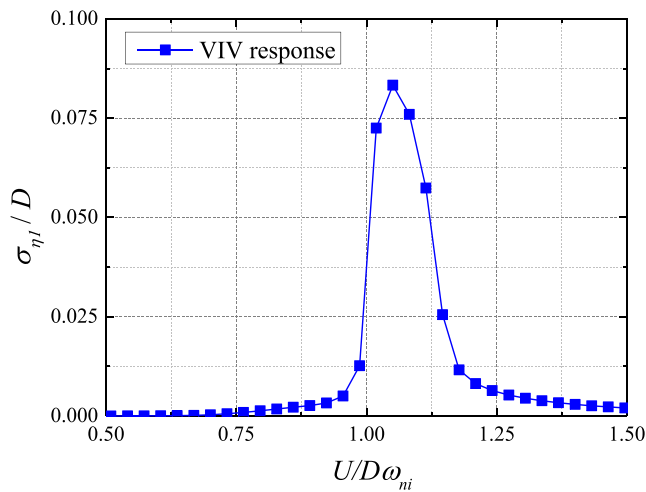


FIGURE 4 Vortex-induced vibration (VIV) response of the deck

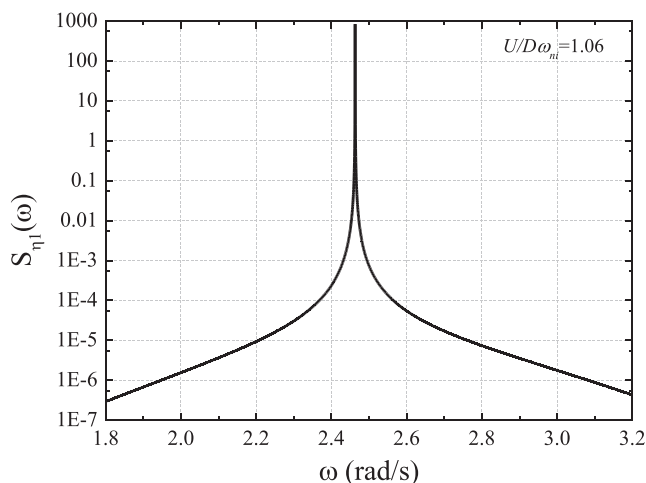


FIGURE 5 Power spectral density of the deck displacement ($U/D\omega_{ni} = 1.06$)

adding into the device, the performances of these devices are significantly improved compared with that of the conventional TMD system.

The optimum values of γ , ν , and ξ_{TMD} can be obtained through the optimization strategy discussed above, and the values are summarized in Table 2. For comparison, the optimum values for the TMD system are also calculated, which can be obtained by setting δ in IBD1 equaling to 0. As mentioned above, IBD1 in the present study is the TMDI arrangement in Hu and Chen,²⁰ and the optimal values can be explicitly calculated by the following equations as suggested by Hu and Chen²⁰:

$$\gamma_{\text{opt}} = \sqrt{\frac{2\delta(1+\mu) + (\mu+2)}{2(1+\mu)^2}}, \quad (19)$$

$$\xi_{\text{opt}} = \sqrt{\frac{1 + \delta^2 + \gamma_{\text{opt}}^4(1+\mu)^2 - \gamma_{\text{opt}}^2(2+\mu) + 2\delta - 2\delta\gamma_{\text{opt}}^2(1+\mu)}{4\gamma_{\text{opt}}^2(1+\mu)}}. \quad (20)$$

In Table 2, the values calculated according to the above formulas are also presented and shown in the brackets. It can be seen that the results obtained by using the proposed optimization method in the present study are almost the same with those obtained by using the analytical equations, which validates the accuracy of the optimization method as adopted in the present study.

Figure 6a shows the spectra of deck response with control devices when $\mu = 0.005$ and $\delta = 2.5$. Comparing with the uncontrolled case in Figure 5, it can be seen that the response of the main deck is significantly reduced with the help of control devices. The magnitude of the PSD of IBD1 is slightly larger than those of the other control devices, which indicates that the control efficiency of IBD1 is not as good as that of others. This is actually expected because the inclusion of the inerter device in parallel with the damping and spring will further resist the free oscillation of the mass block as compared with a conventional TMD (Figure 1a). Therefore, the stroke of the mass block is reduced (as will be shown in the following section), and less energy is dissipated by the damping device. Further comparing the other devices, it can be seen from the zoomed-in chart in Figure 6a that the magnitude of IBD3 is the smallest. The spectrum of IBD4 is slightly larger than that of IBD3 but smaller than that of TMD. As for IBD2, its spectrum is almost identical with that of TMD. According to the optimum design parameters of IBD2 in Table 2 and Equation (15), the magnitude of b in IBD2 is around 10 times larger than C_{TMD} ; thus, the value of the impedance function $Z_2(s)$ (see Table 1) is close to C_{TMD} , which is identical with the impedance of the TMD system. The control efficiency of IBD2 is thereby almost identical with that of TMD. The influence of inertance ratio on the control performance of IBDs will be further analyzed in the following section.

TABLE 2 Optimum design parameters of different IBDs and conventional TMD system ($U/D\omega_{ni} = 1.06$)

Mass parameters	Damper type	γ	ν	ξ_{TMD}
$\mu = 0.005; \delta = 2.5$	IBD1 (TMDI)	1.8656 (1.8655)	/	0.0353 (0.0354)
	IBD2	0.9973	/	0.0353
	IBD3	0.9671	0.2405	0.0617
	IBD4	0.9923	0.0630	0.0352
$\mu = 0.005; \delta = 0$	Conventional TMD	0.9963 (0.9964)	/	0.0353 (0.0354)
$\mu = 0.05; \delta = 2.5$	IBD1 (TMDI)	1.8195 (1.8196)	/	0.1093 (0.1092)
	IBD2	0.9734	/	0.1096
	IBD3	0.9017	0.4425	0.1613
	IBD4	0.9635	0.0905	0.1086
$\mu = 0.05; \delta = 0$	Conventional TMD	0.9642 (0.9641)	/	0.1098 (0.1095)

Abbreviations: IBD, inerter-based damper; TMD, tuned mass damper; TMDI, tuned mass damper inerter.

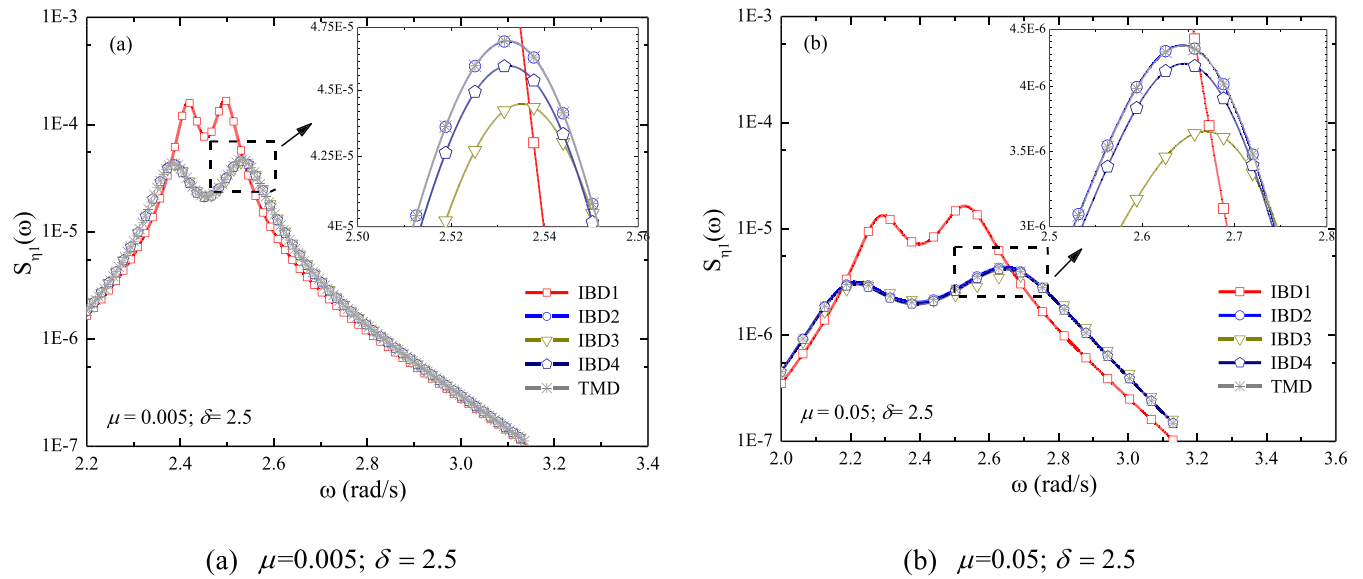


FIGURE 6 Power spectral densities of the main deck displacement with different control devices ($U/D\omega_{ni} = 1.06$). IBD, inerter-based damper; TMD, tuned mass damper

When the mass ratio increases from $\mu = 0.005$ to $\mu = 0.05$, as shown by Figure 6b, the magnitude of these spectra obviously decreases, which indicates that the control efficiency of the IBD systems increases with the increment of mass ratio. This is expected because these IBD systems in essence are TMD systems, in which larger mass ratio results in better control performance.

The above results (Figure 6) lead to the same trend of the normalized RMSs of the deck response (with respect to deck height) as shown in Figure 7, from which it can be seen that IBD1 results in the largest deck response, and the control performances of TMD and IBD2 are almost similar, which is larger than IBD4, whereas IBD3 leads to the best control effectiveness. However, it should be noted that even for IBD1, the maximum deck response is reduced from 0.085 (the uncontrolled case, as shown in Figure 4) to 0.00185, which is less than 3% of the uncontrolled deck response. When the mass ratio μ increases from 0.005 to 0.05, the control performance further improves as shown in Figure 7b.

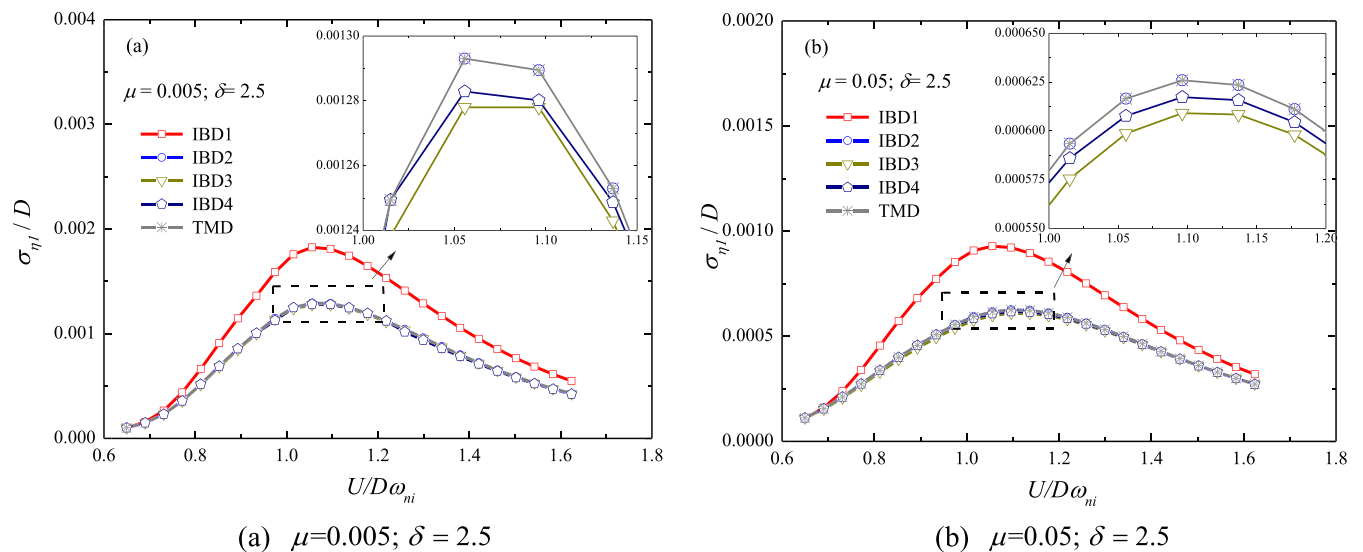


FIGURE 7 Normalized root mean squares of the deck response at different wind velocities. IBD, inerter-based damper; TMD, tuned mass damper

3.3 | Static deformations and strokes of different IBDs

The static deformation and stroke of IBD are important for bridge VIV control, especially for the streamlined box deck where the internal space is limited. The static deformation of IBDs due to gravity can be assessed by the following equation as discussed in Section 1:

$$x_{\text{static}} = g/\omega_{\text{TMD}}^2 = 9.8/(\gamma\omega_{ni})^2. \quad (21)$$

The static deformations of different control devices are calculated, and they are shown in Figure 8. It can be seen that the static deformation of IBD1, which is around 0.5 m, is much smaller than those of others (which are all more than 1.6 m). Considering that the deck height of the example bridge is only 2.5 m, this capacity makes IBD1 more competitive than the other control devices. The static deformations of IBD2 and IBD4 are similar to that of TMD, whereas the static deformation of IBD3 is slightly larger than that of TMD. This is because, as shown in Table 2, the optimal γ values for IBD2, IBD4, and TMD are similar, and they are larger than that in IBD3. Larger γ results in smaller static deformation as shown in Equation (21). When the mass ratio u increases from 0.005 to 0.05, the static deformations of these IBDs also increase, which indicates that the space requirement of these IBDs increase with the increase of mass ratio. This again can be explained by the optimal γ values as shown in Table 2.

Figure 9 shows the stroke of the mass block of each IBD. It can be seen that the stroke of IBD1 is the smallest. This is because, as discussed above, the inclusion of the inerter device in parallel with the damping and spring of a conventional TMD further resists the free oscillation of the mass block. Because the oscillation of IBD1 is constrained, the stroke of this device is thereby reduced. As for the other devices, the stroke of IBD4 is slightly smaller than those of IBD2 and TMD, whereas IBD3 leads to the largest stroke. This is because, as shown in Figure 1, the damping is in series with the inerter and additional spring K_1 in IBD3. The stroke of IBD3 (relative deformation between its two nodes) consists of the deformations of all these three elements. Thus, for the same level of energy dissipation (i.e., deformation of damping device), the stroke of IBD3 will be larger than that of TMD. For IBD4, the additional spring K_1 is in parallel with the damping device, which are further in series with the inerter as shown in Figure 1; the inerter in the IBD4 system thus acts like an additional mass to the conventional TMD mass block (m_{TMD} in Figure 1), which has a “mass amplification” effect on the conventional TMD system. Larger TMD mass results in better control effectiveness as discussed. The control efficiency of this system is thus better, and the stroke is smaller than that of the TMD system with the same mass ratio.

Adding the static deformation (Figure 8) and the corresponding stroke (Figure 9) together, the space requirement for different IBD systems can be obtained. It is obvious that the space requirement of IBD1 is much smaller than that of other IBDs, and the space requirements for IBD4 and IBD2 are slightly smaller and almost identical with that of the conventional TMD. However, for IBD3, the static deformation and stroke are larger than others, and this device thus requires the largest space to be implemented.

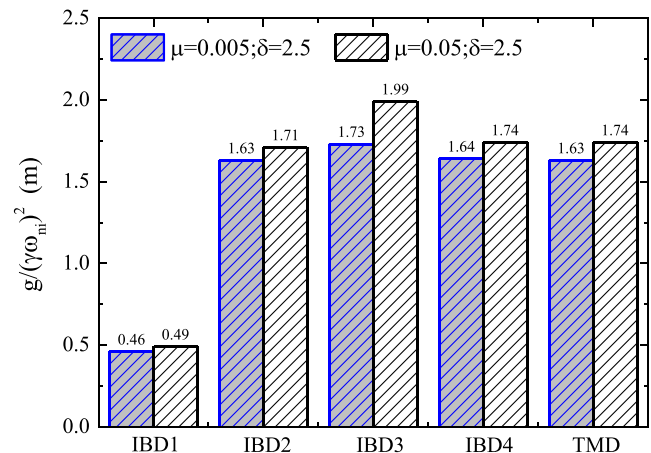


FIGURE 8 Static deformations of the inerter-based damper (IBD) springs due to gravity

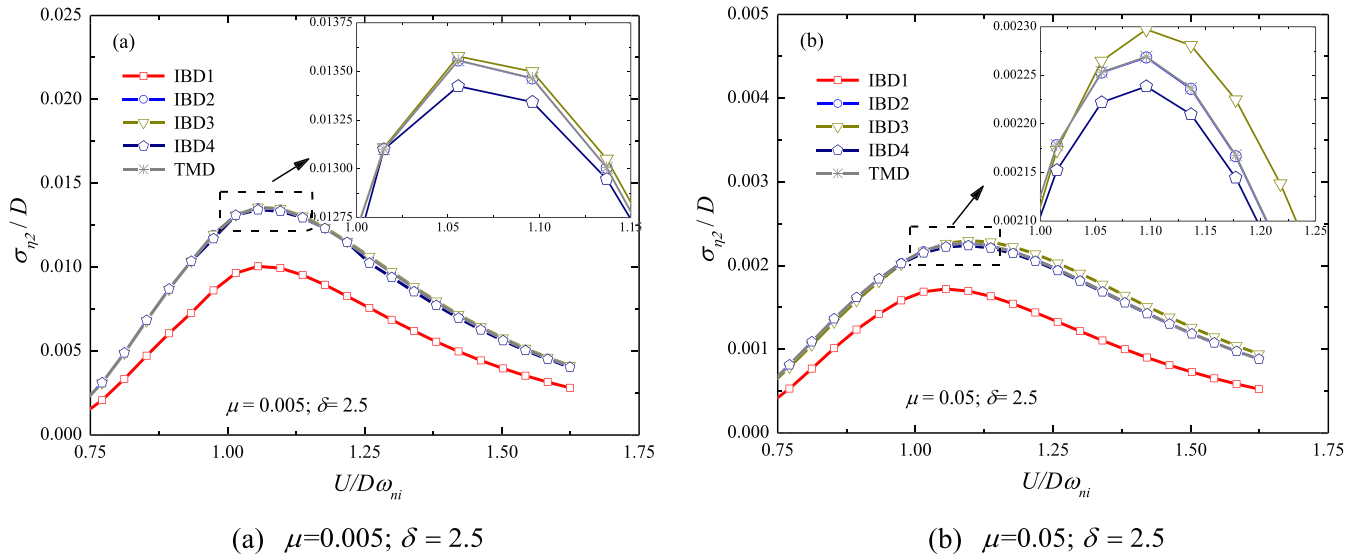


FIGURE 9 Strokes of the mass blocks in different IBDs. IBD, inerter-based damper; TMD, tuned mass damper

3.4 | Reaction forces of IBDs on the bridge deck

In the design process, it is important to foresee the reaction forces associated with the IBDs on the bridge deck because the higher reaction force implies that a strong connection between the device and bridge deck might be necessary.

According to Equation (4), the reaction force of each IBD on the bridge deck can be estimated by

$$\sigma_{F_{IBD}} = |K_{TMD} + j\omega_{TMD}Z(j\omega_{TMD})| \cdot \sigma_{\eta_2}, \quad (22)$$

in which $\sigma_{F_{IBD}}$ is the RMS of the IBD reaction force on the deck, the symbol $|\cdot|$ denotes the module of the complex impedance function, and σ_{η_2} is the stroke of the mass block obtained in the above section. Combining Equation (22) with $Z(j\omega_{TMD})$ in Table 1 and Equations (15) and (16) and taking the optimum design parameters of IBDs in Table 2, the RMS of IBD reactions on the deck can be obtained.

Figure 10 compares the RMS of the IBD reaction forces on the deck at the most unfavorable wind velocity ($U/D\omega_{ni} = 1.06$), where the stroke of mass block (as well as the IBD reaction force on the deck) is the maximum. It can be seen that IBD1 yields the highest reaction force in comparison with the other control scenarios. This is because the inerter device in IBD1 is directly added between the mass block and the bridge deck. For this arrangement, the energy stored in the inerter will transfer back to the structure when the mass block is approaching the maximum displacement and tending to move slower. At these time intervals, the inerter device no longer resists but drives the motion of the mass block due to the accumulated angular momentum in it, which will transfer considerable forces to the support

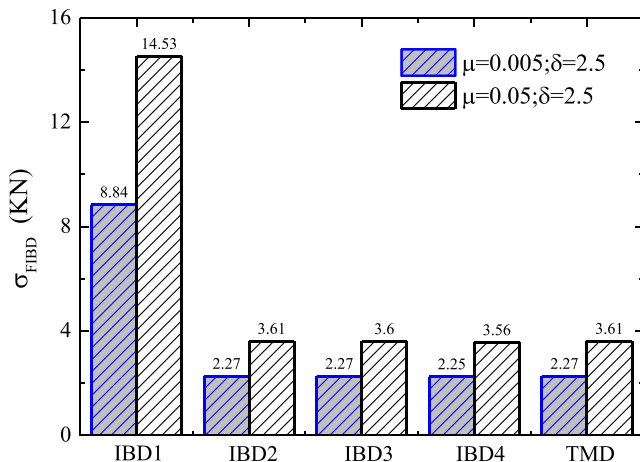


FIGURE 10 Reaction forces of different IBDs on the bridge deck ($U/D\omega_{ni} = 1.06$). IBD, inerter-based damper; TMD, tuned mass damper

ends of the inerter device. This phenomenon was already highlighted by Makris and Kampas¹³ and Málaga-Chuquitaype et al.¹⁴ To overcome this problem, the concept of clutched inerters, as proposed by Makris and Kampas,¹³ might be adopted. The clutch can disengage the rotating flywheel from the structure when the mass block tends to reverse after attaining a local maximum displacement. In this scenario, the inerter can only resist the motion of the mass block but will not drive it. As for the other IBD systems (IBD2 to IBD4), the reaction forces are similar to those of the TMD system.

3.5 | Robustness of different systems

The robustness of the bridge-IBD systems is another important index for evaluating the performance of IBD because some uncertainties unavoidably exist in real engineering practices. For example, it is difficult to get the exact vibration characteristics of the main structure, and some inaccuracies might occur when the damping and/or spring elements are manufactured. The robustness of different IBD systems are investigated through a sensitivity analysis in this subsection.

Figure 11 shows the contour lines of control efficiency of IBDs against different design parameters (ξ_{TMD} and γ) when $\mu = 0.005$ and $\delta = 2.5$. In these figures, the control efficiency is defined as $1 - \sigma_{\eta_1} / \sigma_{\eta_1 - \text{uncontrol}}$, which denotes the ratio of response reduction of the bridge deck. For comparison, the control results under the TMD system are also plotted. It can be seen that the shapes of the contour lines of the TMD (Figure 11a) and IBD1 (Figure 11b) systems are similar, which indicates that the robustness of these two systems are similar to each other. However, the values of contour lines of IBD1 are slightly smaller than those of TMD. This is because, as discussed above, the control efficiency of IBD1 is slightly worse than TMD.

The contour lines of IBD2 (Figure 11c) with respect to frequency ratio are slightly larger than those of TMD and IBD1 especially at low damping ratios, which means the robustness of IBD2 system is better than those of TMD and IBD1. Namely, through combining the inerter device in series with the damper of a conventional TMD system (i.e., the IBD2 network), although the control efficiency keeps almost the same (refers to Figures 6 and 7), the robustness of the system can be further improved. For IBD3 (Figure 11d), the contour lines show a narrow-banded property with respect to frequency ratio, which indicates the control efficiency of this system is quite sensitive to the frequency ratio. Once the stiffness of the spring slightly deviates from the optimum value, the control efficiency of this system will dramatically decrease. In this sense, this system is not robust. As for IBD4, the contour lines are much wider than that of other systems. As shown in Figure 11e, the control efficiency is larger than 0.9 in most regions, which means this system is quite robust and the control effectiveness is not sensitive to the parameters of the system. This will significantly facilitate the design of the system.

Figure 12 shows the results when $\mu = 0.05$ and $\delta = 2.5$. Comparing with Figure 11, the contour lines become much wider against the design parameters, which means increasing the mass ratio u can increase the robustness of the control system. This is again consistent with the conventional TMD system. Similar with the above results, IBD4 is the most robust design, whereas IBD3 results in the worst robustness.

4 | FURTHER DISCUSSIONS

4.1 | Influence of mass ratio and inertance ratio

All the above results are based on the assumed mass ratio u (0.005 and 0.05) and inertance ratio δ (2.5). To further examine the influences of both parameters, the performances of different IBDs with a variety of mass and inertance ratios are investigated in this subsection.

Figure 13a shows the normalized deck response (the ratio of the controlled deck response to that of the uncontrolled case) when IBD1 is applied. It can be seen that, at a specific mass ratio u , the normalized deck response increases with the increment of inertance ratio, indicating that the control efficiency of IBD1 decreases with the increasing of inertance ratio. This is because, as discussed above, the inclusion of the inerter actually prevents the free oscillation of the conventional TMD system (where only the spring and damper are included). The control effectiveness is thus slightly decreased. However, it is noticeable that even for the most unfavorable condition (i.e., $\mu = 0.005$ and $\delta = 10$), the controlled deck response is still less than 3% of the uncontrolled value. On the other hand, increasing the mass ratio

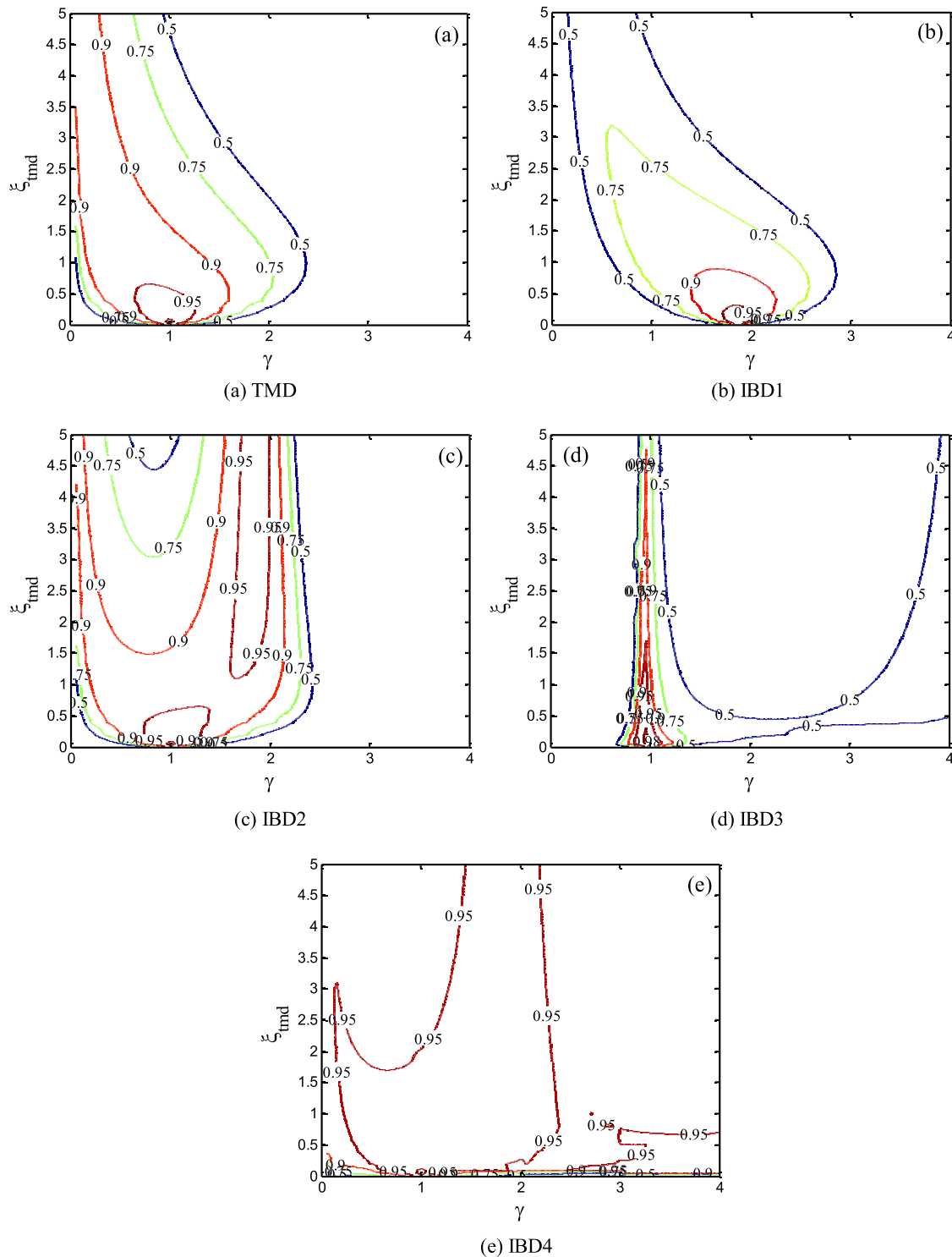


FIGURE 11 Influence of design parameters on the control efficiencies of different IBDs ($\mu = 0.005$ and $\delta = 2.5$). IBD, inerter-based damper; TMD, tuned mass damper

u , the control effectiveness becomes more evident, which is consistent with our understanding on the TMD system. As to the stroke of mass block as shown in Figure 13b, increasing the inertance ratio slightly decreases the stroke when the mass ratio u is fixed. This is again because the free oscillation of the mass block is prevented by the inerter.

Figure 13c,d shows the optimum parameters of IBD1. It is noticeable that the value of γ increases with the increment of inertance ratio (Figure 13c). According to Equation (21), the static deformation of the spring is inversely

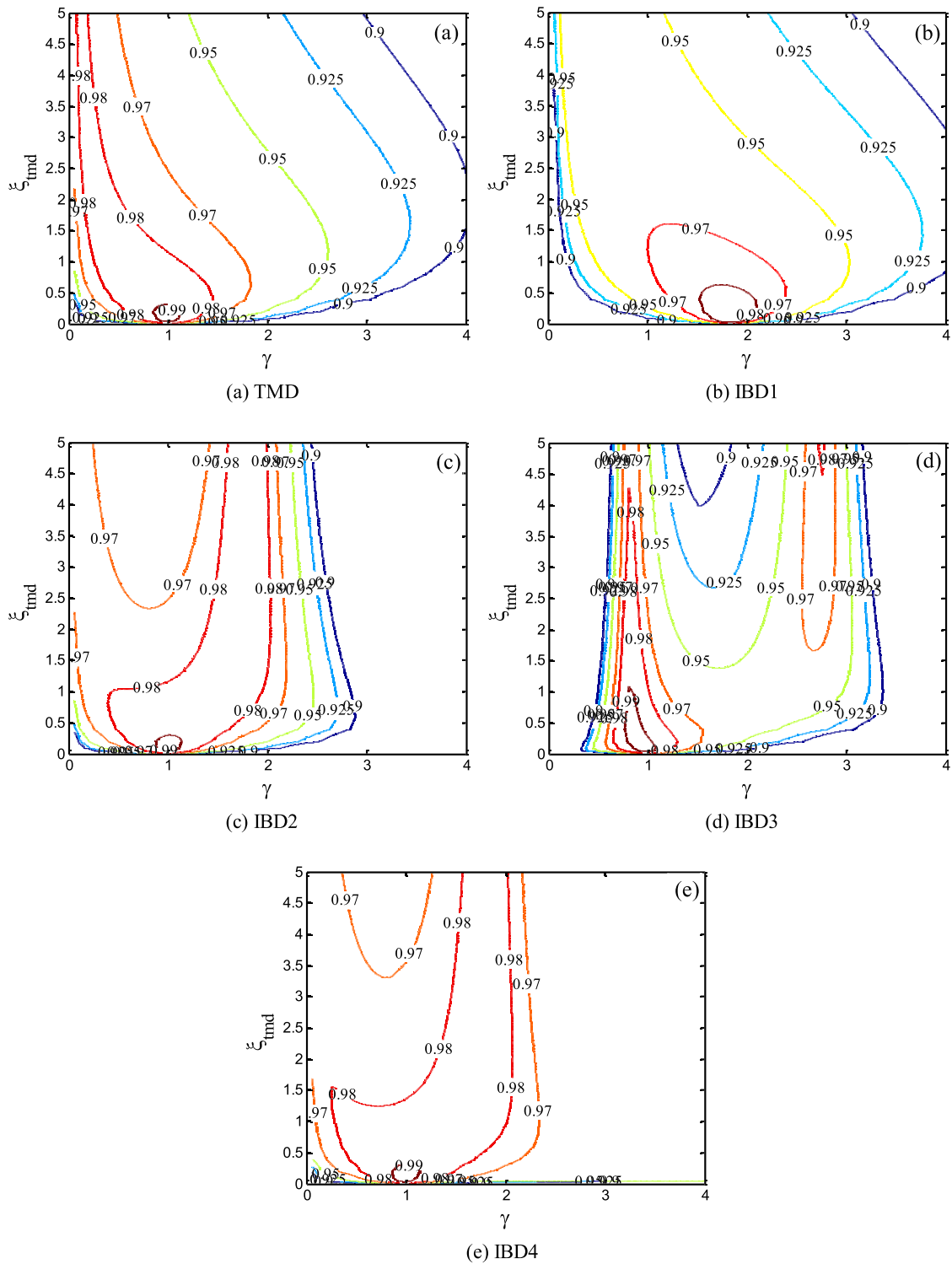


FIGURE 12 Influence of design parameters on the control efficiencies of different IBDs ($\mu = 0.05$ and $\delta = 2.5$). IBD, inerter based damper; TMD, tuned mass damper

proportional to the square of γ . In this sense, the static deformation of IBD1 can be obviously reduced by increasing the inertance ratio. As for the damping ratio (Figure 13d), it can be seen that it is not sensitive to the inertance ratio.

Figure 13e shows the reaction force of IBD1 on the bridge deck. As can be seen, it slightly decreases with the increment of inertance ratio when $\delta \leq 1$. After that, the reaction force dramatically increases with the increment of inertance ratio. This is because the reaction force induced by the inerter device is antiphase with that of the spring in IBD1 (see the impedance function of IBD1). When the inertance ratio is small, the spring dominates the reaction force of this

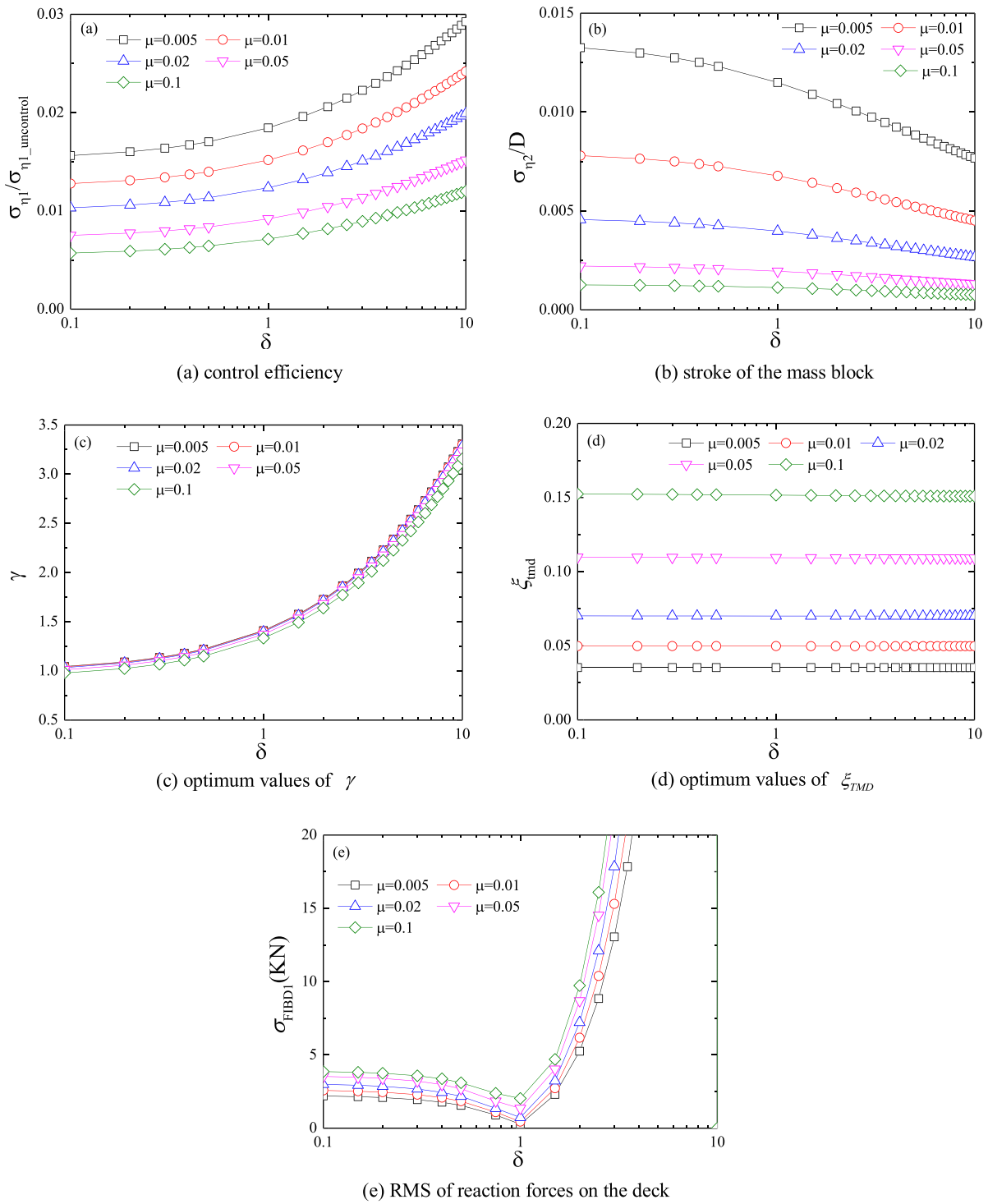


FIGURE 13 Influences of mass and inertia ratios on the control performance, stroke of the mass block, optimal design parameters, and reaction forces on the deck (IBD1). IBD, inerter-based damper; RMS, root mean square

system, and the increment of inertia ratio reduces the module of the impedance function and thus reduces the reaction force on the deck. However, with the increment of inertia ratio, the inerter device dominates, and the reaction force increases with the increment of inertia ratio. For IBD1 with large values of inertia ratio (e.g., $\delta \geq 5$), the reaction force on the deck becomes quite large in comparison with conventional TMDs. Because the increase of inertia ratio can reduce the static deformation of the springs but will lead to the reduction of control efficiency and increment of reaction force on the deck, the selection of inertia value for IBD1 is quite important. It should be predetermined on the basis of an overall consideration of the application situation and the pros and cons of this device. For this

example bridge, an inertance ratio of $\delta \leq 2.5$ is recommended. If $\delta = 2$ is chosen, the static deformation of springs is around one third of TMDs, and the reaction force on the bridge deck is around three times of TMDs.

Figure 14a,b shows the control efficiency and stroke of IBD2, respectively. For comparison, the results for the TMD case are also plotted. It can be seen that, at a specific mass ratio μ , the normalized deck response and the stroke gradually decrease with the increment of inertance ratio, and they finally equal to the corresponding values of the TMD system with the same mass ratio. This indicates that the control efficiency of IBD2 is worse than that of conventional TMD at small values of δ . The influence of inerter device on the control efficiency of IBD2 can

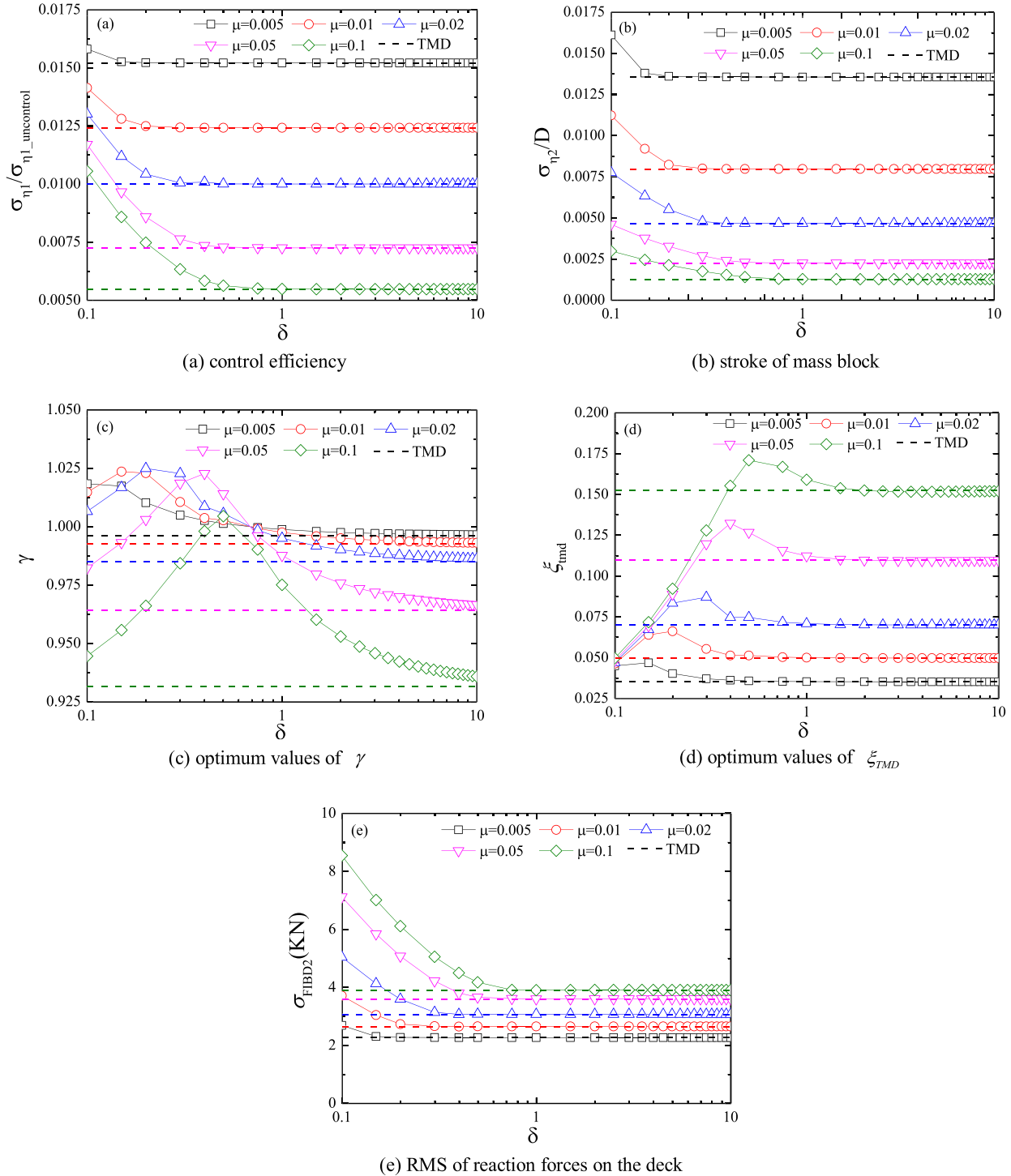


FIGURE 14 Influences of mass and inertance ratios on the control performance, stroke of the mass block, optimal design parameters, and reaction forces on the deck (IBD2). IBD, inerter-based damper; RMS, root mean square; TMD, tuned mass damper

be explained by the impedance function of $Z_2(s)$ in Table 1. At small values of δ (i.e., inertance b), $1/(bs)$ in $Z_2(s)$ is large, and the impedance function $Z_2(s)$ is smaller than C_{TMD} . With the increment of b , $1/(bs)$ in $Z_2(s)$ approaches zero, and the impedance function $Z_2(s)$ is thereby close to C_{TMD} , which is identical with the impedance of a conventional TMD system. The control efficiency and stroke of IBD2 are then almost identical with those of TMD.

Figure 14c,d shows the optimum design parameters of IBD2. It can be seen that the optimal frequency ratio γ increases initially and then gradually decreases with the increment of δ , and the optimal frequency ratio of IBD2 is always larger than that of the corresponding TMD system, which indicates that the static deformation of spring in IBD2 is smaller than that of TMD. As for the optimum damping ratios, they initially increase and then slightly decrease with the increment of δ . When δ is relatively large, the optimum damping ratio is almost identical with that of TMD at the same mass ratio.

Figure 14e shows the reaction force of IBD2 on the bridge deck. It can be seen that, for a specific u , the reaction force gradually decreases with the increment of inertance ratio and finally equals to the corresponding value of the TMD system with the same mass ratio. As discussed above, this is because the impedance function of the IBD2 system gradually approaches that of TMD system with the increment of inertance ratio.

Figure 15 shows the results of IBD3. It can be seen from Figure 15a that, different from IBD2, the normalized deck response slightly increases with the increment of inertance ratio, and this phenomenon becomes more obvious when the mass ratio u is large. Moreover, the values of the normalized deck responses are smaller than those of the TMD system at same mass ratios, which indicates that IBD3 has a better control performance compared with TMD. As to the stroke of the mass block (Figure 15b), the difference between IBD3 and TMD is not obvious. Figure 15c shows the optimum frequency ratios of IBD3. For a specific mass ratio u , the optimum frequency ratio gradually decreases with the increment of inertance ratio except in the case with $u = 0.1$, in which it can be seen that when δ is quite small, γ actually increases slightly. Moreover, it can be seen that the optimum frequency ratios are smaller than the corresponding values of TMD in most cases. According to Equation (21), a smaller γ denotes a larger static deformation of the spring. On the basis of this point and considering the robustness of this system, IBD3 may not be applicable for bridge VIV control, even though its control efficiency is better than TMD with the same mass ratio. The optimum damping ratio of IBD3 (Figure 15d) gradually increases with the increment of inertance ratio, the value of which is slightly larger than that of TMD with the same mass ratio. Because an additional spring K_1 is included in IBD3, the corner frequency ratio ν should also be optimized, and Figure 15e shows the results. It can be seen that the optimum corner frequency ratio gradually decreases with the increment of inertance ratio. As for the reaction force, it is close to that when the bridge is controlled by a TMD system with the same mass ratio (Figure 15f).

Figure 16 shows the results of IBD4. Similar with IBD3, the normalized deck response slightly increases with the increment of inertance ratio. However, different from IBD3, this trend is more obvious when the mass ratio u is small (Figure 16a). The normalized deck responses of IBD4 are smaller than those of TMD with the same mass ratio. Thus, the control efficiency of this system is better than that of TMD. As for the stroke of IBD4, Figure 16b shows that it also slightly increases with the increment of inertance ratio, and this trend is more obvious at small mass ratio u .

Figure 16c shows the optimum frequency ratios of IBD4. It is noticeable that at a specific mass ratio u , the optimum frequency ratio of IBD4 initially decreases and then gradually increases with the increment of inertance ratio. In particular, when the inertance ratio is small, the optimum frequency ratio is smaller than the corresponding TMD system. However, when the inertance ratio is relatively large, the optimum frequency ratio becomes larger than that of TMD system. As to the optimum damping ratio (Figure 16d), it gradually increases and finally approaches the corresponding value of the TMD system with the same mass ratio. The optimum corner frequency ratio gradually decreases with the increment of inertance ratio (Figure 16e). This trend is similar to that of IBD3, but the decrease rate is more obvious than that of IBD3. As for the reaction force (Figure 16f), it is slightly larger than that of TMDs for $u = 0.02$, $u = 0.05$, and $u = 0.1$ in the range of small inertance ratios ($\delta \leq 1$). For $u \leq 0.01$, the reaction force of IBD4 is almost identical with that of the TMD system.

From the above results, the property of IBD4 is adjustable on the basis of the applicable conditions of VIV control. For example, if a small inertance ratio (e.g., $\delta < 0.5$) is selected, the control efficiency of this system is better than that of the TMD system with the same mass ratio, and the stroke is also reduced. However, the static deformation of spring and the reaction force will be slightly larger than those of TMDs. On the other hand, if a large inertance ratio (e.g., $\delta > 4$) is chosen, the static deformation of the spring is obviously smaller than that of the TMD system with the

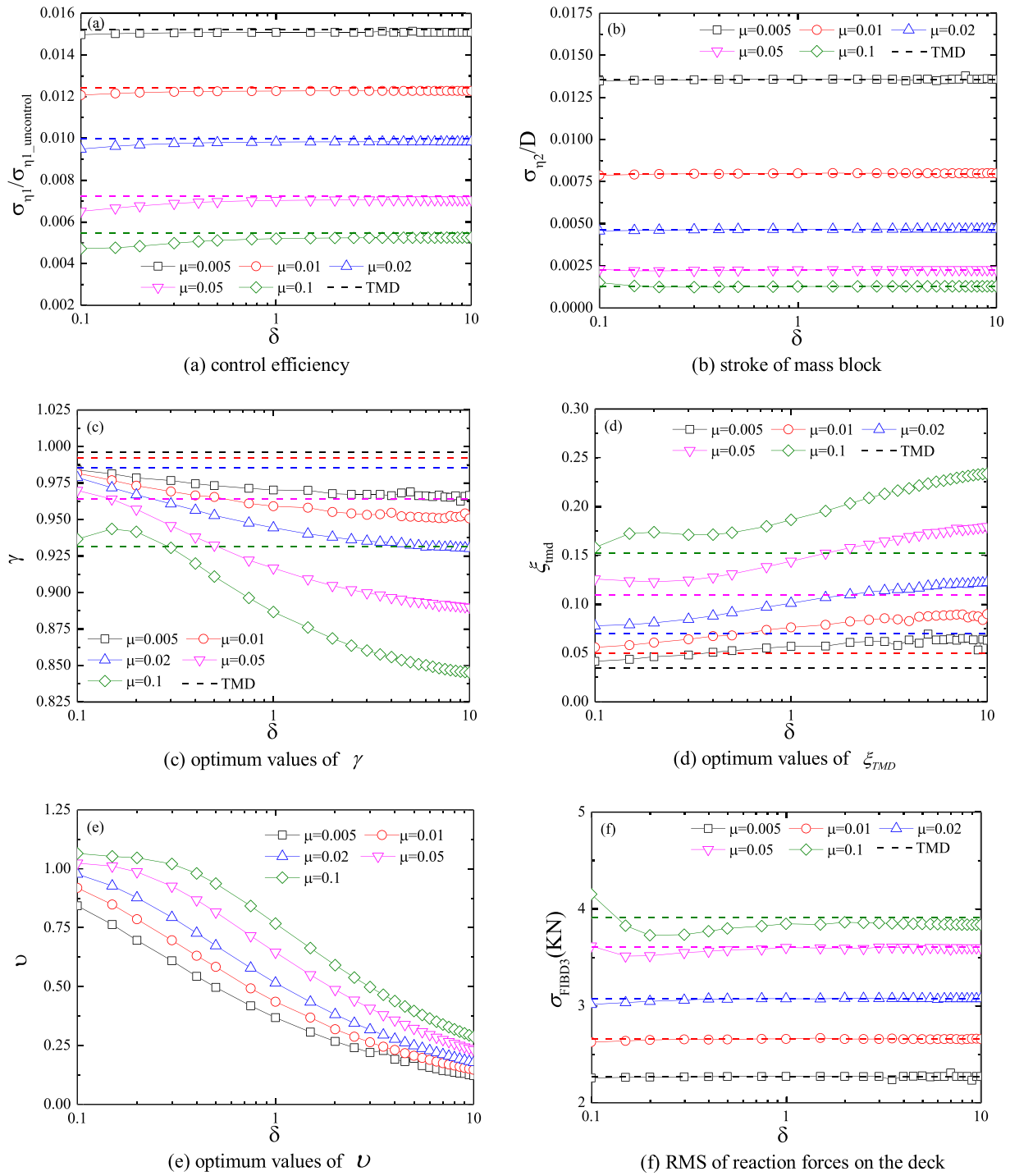


FIGURE 15 Influences of mass and inertia ratios on the control performance, stroke of the mass block, optimal design parameters, and reaction forces on the deck (IBD3). IBD, inerter-based damper; RMS, root mean square

same mass ratio, whereas the stroke and reaction force are almost identical with those of the TMD system. The control efficiency is also improved as compared with TMD, but the extent of improvement is not as obvious at small inertia ratios.

Considering the control efficiency and robustness of this system, IBD4 is an appealing option for bridge VIV control. It requires similar applicable conditions with conventional TMDs (magnitudes of damping, stiffness, connection elements, and installation space) but has a better control efficiency and robust performance. In this sense, it can be adopted to replace conventional TMDs in bridge VIV control.

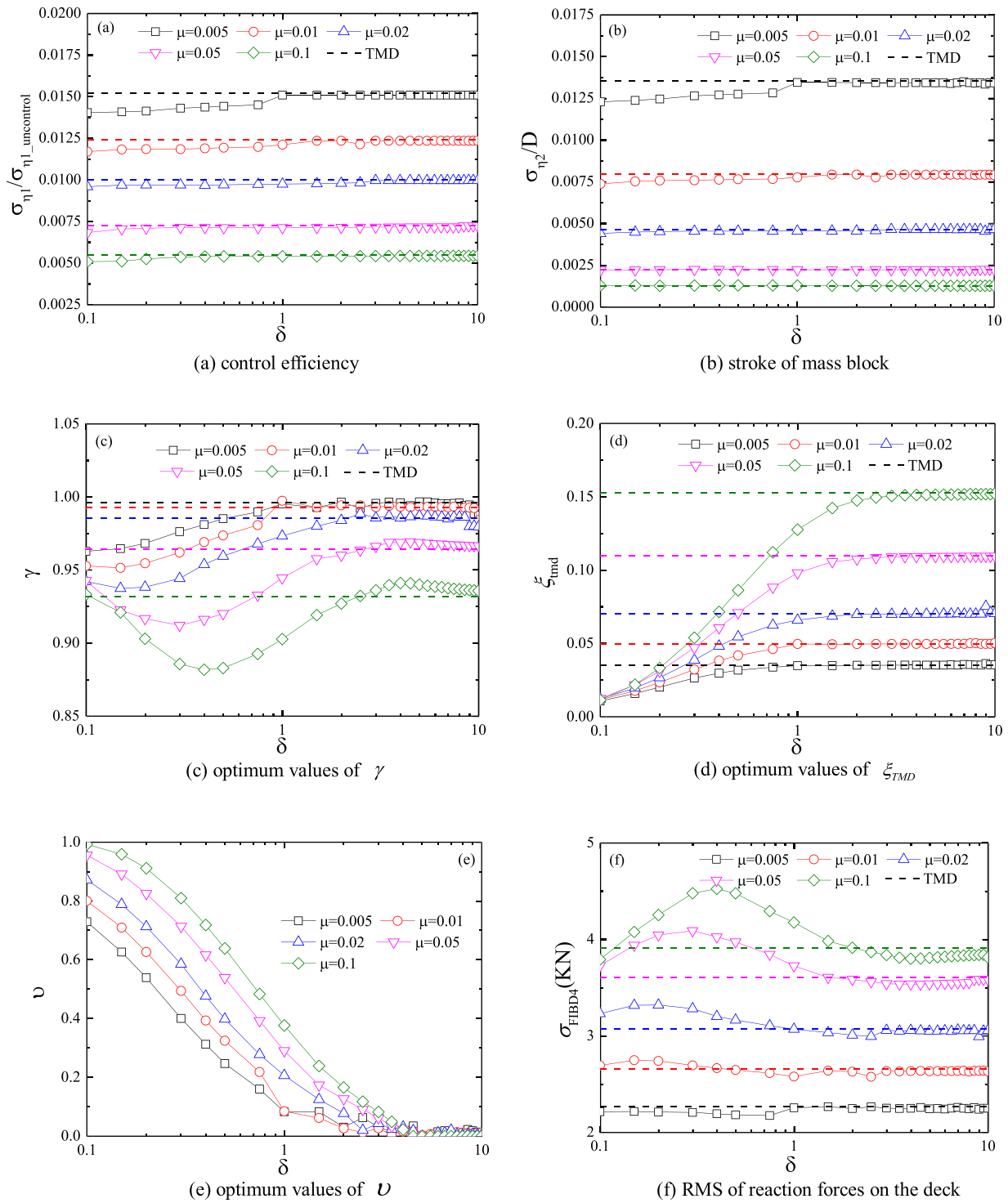


FIGURE 16 Influences of mass and inertia ratios on the control performance, stroke of the mass block, optimal design parameters, and reaction forces on the deck (IBD4). IBD, inerter-based damper; RMS, root mean square; TMD, tuned mass damper

4.2 | Control efficiency at optimum inertia ratios

In the design of the conventional TMD system, the mass ratio u is normally predefined. Following the same principle, the inertia ratio δ is also determined in advance in the above sections. Actually, the inertia ratio can also be taken as a parameter to be optimized in the design. According to the above results, IBD1 slightly decreases the control efficiency compared with the conventional TMD system, and the control effectiveness of IBD2 is almost the same as that

of conventional TMD. The optimum values of inertance ratio for these two systems are thus zero. Therefore, only the optimum inertance ratios and performances of IBD3 and IBD4 are obtained and compared with TMD in this section.

Figure 17a shows the optimum inertance ratio δ at different mass ratios μ ; it can be seen that δ gradually increases with the increment of mass ratio μ . For all the considered mass ratios, the optimum inertance ratios are less than unity, which means if an inerter device that can amplify its physical mass by 100 times is adopted, the physical mass of this device will be less than 1% of the TMD mass block. Because the required amount of inertance in these IBDs is quite small and the reaction forces of these two IBD systems are similar to that of conventional TMDs, the increasing of the manufacturing cost is therefore not obvious. Figure 17b shows the optimum frequency ratios. It can be seen that γ gradually decreases with the increment of the mass ratio. For a particular mass ratio, the optimum frequency of IBD4 is close to that of TMD and is smaller than that of IBD3. In this sense, the static deformation of IBD4 is similar to that of TMD, whereas the static deformation of IBD3 is slightly smaller than that of the others. Figure 17c shows the optimum damping ratios. As shown, it gradually increases with the increment of mass ratio. For a particular mass ratio, the optimum damping of IBD4 is the smallest, whereas the optimum damping of IBD3 is similar to that of TMD. Figure 17d shows the optimum corner frequency ratios, which determine the stiffness of the additional springs in IBDs (K_1 in Figure 1). It can be seen that the optimum corner frequency ratios are around 1. According to Equations (15) and (16), the corner frequency ratio can be expressed as $\nu = \omega_b/\omega_{TMD} = \sqrt{K_1/(\delta K_{TMD})}$. Because the values of optimum inertance ratios are smaller than 1 (Figure 17a), the required amount of additional stiffness K_1 in IBDs will be smaller than the TMD spring K_{TMD} .

Figure 18 shows the control efficiency and stroke of the IBD3 and IBD4. As can be seen, the control efficiency gradually increases with the increment of the mass ratio. For a particular mass ratio, the control efficiency of IBD3 is the best among these three devices. The control efficiency of IBD4 is close to that of IBD3 at small mass ratios (less than 0.1) and gradually approaches to that of TMD at large mass ratios (larger than 0.25). Because the mass ratio of IBDs for

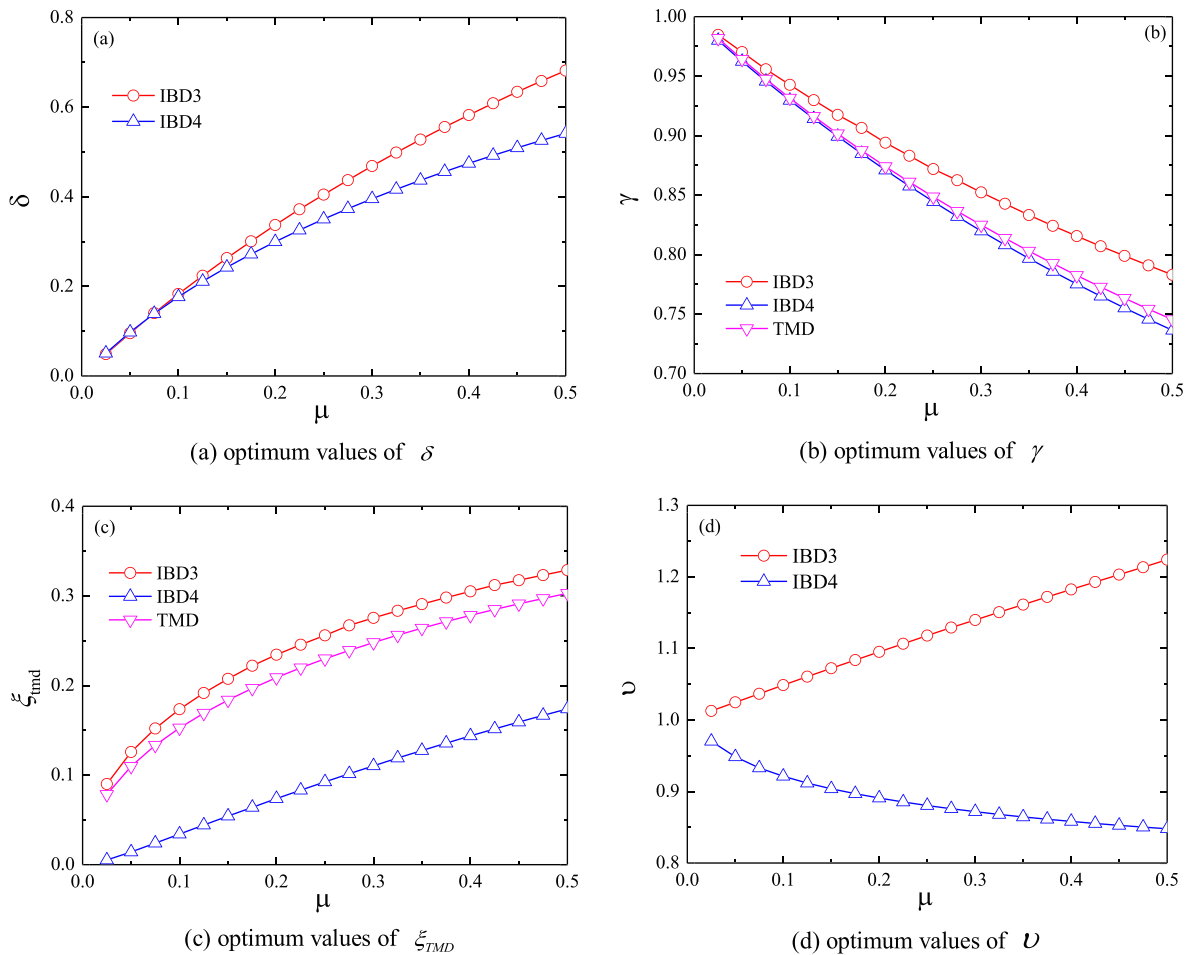


FIGURE 17 Optimum design parameters of IBD3 and IBD4. IBD, inerter-based damper; TMD, tuned mass damper

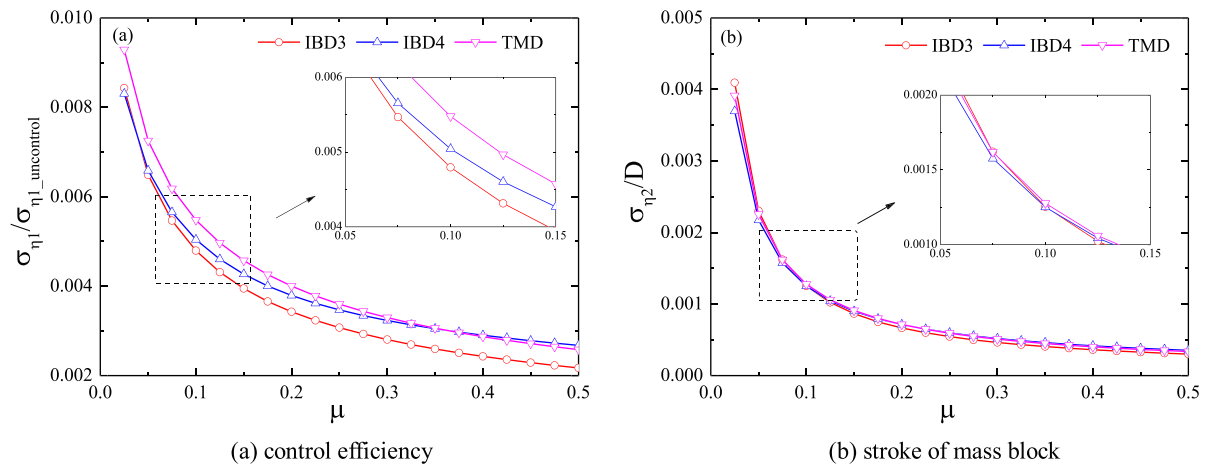


FIGURE 18 Control efficiency of IBD3 and IBD4 at optimum inertance ratios. IBD, inerter-based damper; RMS, root mean square; TMD, tuned mass damper

bridge VIV control normally will not exceed 0.25, it is thus appropriate to conclude that the control efficiency of IBD4 is better than that of TMD.

To further illustrate the efficiency of IBD3 and IBD4, the normalized deck response of $\sigma_{\eta_1}/\sigma_{\eta_1-uncontrol} = 0.005$ is taken as an illustration (see the zoomed-in chart in Figure 18a). It can be seen from the figure that the mass ratio that corresponds to $\sigma_{\eta_1}/\sigma_{\eta_1-uncontrol} = 0.005$ is around 0.1 for IBD3 and IBD4, and it is around 0.125 for TMD. In this sense, the auxiliary mass of IBDs is 20% less than that of TMD for the same level of control efficiency. Figure 18b shows the stroke of mass block. For a particular mass ratio, the strokes of IBD3 and IBD4 are slightly smaller than that of TMD. However, the difference between these control devices is not obvious. Thus, the stroke reduction of IBDs can be conservatively neglected.

5 | SUGGESTION FOR VIV-RESISTANCE DESIGN

The above results show that, compared with the conventional TMD, although the control efficiency is slightly reduced when IBD1 is used, it significantly reduces the static deformation of springs and the stroke of the mass block. Thus, this system is quite applicable for long-span bridges with streamlined box decks, where the deck height is often limited and the TMD scheme may be restricted. For IBD4, the installation space requirement is similar to that of conventional TMD. However, its control efficiency and robustness are obviously better. Thus, this system can replace the conventional TMD with same mass ratio to further enhance the control efficiency or reduce the amount of auxiliary mass for the same level of control efficiency. As for IBD3, even though the control efficiency is also better than conventional TMD, it may not be proper for bridge VIV control because the robustness of this system is narrow-banded and is sensitive to the design parameters.

For long-span bridges, IBD1 thus can be adopted to control the low mode VIVs of the bridge, whereas IBD4 can be adopted to control the high mode VIVs of the bridge. Moreover, conventional TMDs are narrow-banded, and VIV control of long-span bridges by using TMDs is often conducted mode by mode. A TMD that is valid for one mode may be invalid for the other modes. Thus, the number of TMDs can be quite large to suppress all potential VIV modes of the bridge. Comparing with conventional TMDs, the broad-banded property of IBD4 is quite appealing. It is expected to achieve a desirable control effect for all potential VIV modes with a limited number of IBD4s through proper distribution span wise along the bridge.

6 | CONCLUSIONS

The application of IBDS for VIV control of long-span bridges is conducted in the present study. The governing equations of four bridge-IBD systems under the effect of VIV are established. The optimal design strategy for the IBD parameters

is proposed. The feasibilities of these IBDs in bridge VIV control are systematically assessed by examining several important indicators, such as the control efficiency, static deformation of springs, stroke of the mass block, reaction force of IBD on the bridge deck, and robustness of the system. The main findings and conclusions of this study are as follows:

1. A combination of an inerter device in parallel with the damping and stiffness of a conventional TMD system (i.e., IBD1) can significantly reduce the static deformation of springs and stroke of the mass block. It is quite suitable to control the VIV of bridges where the deck height is limited and conventional TMD scheme is restricted. However, the reaction force of IBD1 on the bridge deck will be increased in comparison with conventional TMDs. The application of IBD1 and the predetermination of its inertance should be conducted on the basis of an overall consideration of the application situation and the pros and cons of this device.
2. IBDs with additional stiffness (i.e., IBD3 and IBD4 in this study) have a better control efficiency than conventional TMD. Among these IBDs, IBD4 has similar applicable conditions with conventional TMD (i.e., similar space requirement and mounting system), but its control efficiency and robustness are much better.
3. Due to the broad-banded property of IBD4, it is expected to achieve a desirable control effect for all potential VIV modes with a limited number of dampers through proper distribution along the span wise of the bridge.
4. It should be noted that only analytical studies are performed. To further demonstrate the effectiveness of the IBD systems, an in-depth case study with realistic modeling of structure and load (and interactions) and the mechanical properties of the inerter device and their influences on the control efficiency are needed and will be conducted in future studies.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the support of the National Natural Science Foundation of China (51708011, 51778023) and the China Postdoctoral Science Foundation funded project (2018 T110022).

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How to cite this article: Xu K, Bi K, Ge Y, Zhao L, Han Q, Du X. Performance evaluation of inerter-based dampers for vortex-induced vibration control of long-span bridges: A comparative study. *Struct Control Health Monit*. 2020;27:e2529. <https://doi.org/10.1002/stc.2529>

APPENDIX A: CHARACTERISTIC EQUATIONS

1. IBD1:

The characteristic equation of IBD1 is

$$\begin{aligned} & \lambda^4 \left(-\mu^2 M_i^2 + (1 + \mu) M_i (b + \mu M_i) \right) + \lambda^3 \left((1 + \mu) C_{\text{TMD}} M_i + 2 M_i (b + \mu M_i) (-\xi_{ae} + \xi_{ni}) \omega_{ni} \right) \\ & + \lambda^2 \left((1 + \mu) K_{\text{TMD}} M_i + 2 C_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + M_i (b + \mu M_i) \omega_{ni}^2 \right) \\ & + \lambda \left(2 K_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + C_{\text{TMD}} M_i \omega_{ni}^2 \right) + K_{\text{TMD}} M_i \omega_{ni}^2 = 0. \end{aligned} \quad (\text{A1})$$

2. IBD2:

The characteristic equation of IBD2 is

$$\begin{aligned} & \lambda^5 \left[-b\mu^2 M_i^2 + b\mu(1 + \mu) M_i^2 \right] \\ & + \lambda^4 \left[b(1 + \mu) C_{\text{TMD}} M_i - \mu^2 C_{\text{TMD}} M_i^2 + \mu(1 + \mu) C_{\text{TMD}} M_i^2 + 2b\mu M_i^2 (-\xi_{ae} + \xi_{ni}) \omega_{ni} \right] \\ & + \lambda^3 \left[b(1 + \mu) K_{\text{TMD}} M_i + 2b C_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + 2\mu C_{\text{TMD}} M_i^2 (-\xi_{ae} + \xi_{ni}) \omega_{ni} + b\mu M_i^2 \omega_{ni}^2 \right] \\ & + \lambda^2 \left[(1 + \mu) C_{\text{TMD}} K_{\text{TMD}} M_i + 2b K_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + b C_{\text{TMD}} M_i \omega_{ni}^2 + \mu C_{\text{TMD}} M_i^2 \omega_{ni}^2 \right] \\ & + \lambda \left[2 C_{\text{TMD}} K_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + b K_{\text{TMD}} M_i \omega_{ni}^2 \right] \\ & + C_{\text{TMD}} K_{\text{TMD}} M_i \omega_{ni}^2 = 0 \\ & \text{and} \\ & b\lambda + C_{\text{TMD}} \neq 0. \end{aligned} \quad (\text{A2})$$

3. IBD3:

The characteristic equation of IBD3 is

$$\begin{aligned} & \lambda^6 \left[-b\mu^2 C_{\text{TMD}} M_i^2 + b\mu(1 + \mu) C_{\text{TMD}} M_i^2 \right] \\ & + \lambda^5 \left[-b\mu^2 K_1 M_i^2 + b\mu(1 + \mu) K_1 M_i^2 + 2b C_{\text{TMD}} \mu M_i^2 (-\xi_{ae} + \xi_{in}) \omega_{ni} \right] \\ & + \lambda^4 \left[b(1 + \mu) C_{\text{TMD}} K_1 M_i + b(1 + \mu) C_{\text{TMD}} K_{\text{TMD}} M_i - \mu^2 C_{\text{TMD}} K_1 M_i^2 \right. \\ & \quad \left. + \mu(1 + \mu) C_{\text{TMD}} K_1 M_i^2 + 2b\mu K_1 M_i^2 (-\xi_{ae} + \xi_{ni}) \omega_{ni} + b\mu C_{\text{TMD}} M_i^2 \omega_{ni}^2 \right] \\ & + \lambda^3 \left[b(1 + \mu) K_1 K_{\text{TMD}} M_i + 2b C_{\text{TMD}} K_1 M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} \right. \\ & \quad \left. + 2b C_{\text{TMD}} K_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + 2\mu C_{\text{TMD}} K_1 M_i^2 (-\xi_{ae} + \xi_{ni}) \omega_{ni} + b\mu K_1 M_i^2 \omega_{ni}^2 \right] \\ & + \lambda^2 \left[(1 + \mu) C_{\text{TMD}} K_1 K_{\text{TMD}} M_i + 2b K_1 K_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + b C_{\text{TMD}} K_1 M_i \omega_{ni}^2 \right. \\ & \quad \left. + b C_{\text{TMD}} K_{\text{TMD}} M_i \omega_{ni}^2 + \mu C_{\text{TMD}} K_1 M_i^2 \omega_{ni}^2 \right] \\ & + \lambda \left[2 C_{\text{TMD}} K_1 K_{\text{TMD}} M_i (-\xi_{ae} + \xi_{ni}) \omega_{ni} + b K_1 K_{\text{TMD}} M_i \omega_{ni}^2 \right] \\ & + C_{\text{TMD}} K_1 K_{\text{TMD}} M_i \omega_{ni}^2 = 0 \\ & \text{and} \\ & \lambda^2 b + \lambda b K_1 + K_1 C_{\text{TMD}} \neq 0. \end{aligned} \quad (\text{A3})$$

4. IBD4:

The characteristic equation of IBD4 is

$$\begin{aligned}
 & \lambda^6 [-b\mu^2 M_i^2 + b\mu(1+\mu)M_i^2] \\
 & + \lambda^5 [b(1+\mu)C_{TMD}M_i - \mu^2 C_{TMD}M_i^2 + \mu(1+\mu)C_{TMD}M_i^2 + 2b\mu M_i^2(-\xi_{ae} + \xi_{ni})\omega_{ni}] \\
 & + \lambda^4 \left[b(1+\mu)K_1M_i + b(1+\mu)K_{TMD}M_i - \mu^2 K_1M_i^2 + \mu(1+\mu)K_1M_i^2 \right. \\
 & \quad \left. + 2bC_{TMD}M_i^2(-\xi_{ae} + \xi_{ni})\omega_{ni} + b\mu M_i^2\omega_{ni}^2 + 2\mu C_{TMD}M_i^2(-\xi_{ae} + \xi_{ni})\omega_{ni} \right] \\
 & + \lambda^3 \left[(1+\mu)C_{TMD}K_{TMD}M_i + 2bK_1M_i(-\xi_{ae} + \xi_{ni})\omega_{ni} + 2bK_{TMD}M_i(-\xi_{ae} + \xi_{ni})\omega_{ni} \right. \\
 & \quad \left. + 2\mu K_1M_i^2(-\xi_{ae} + \xi_{ni})\omega_{ni} + \mu C_{TMD}M_i^2\omega_{ni}^2 + bC_{TMD}M_i\omega_{ni}^2 \right] \\
 & + \lambda^2 \left[(1+\mu)K_1K_{TMD}M_i + 2C_{TMD}K_{TMD}M_i(-\xi_{ae} + \xi_{ni})\omega_{ni} \right. \\
 & \quad \left. + bK_1M_i\omega_{ni}^2 + bK_{TMD}M_i\omega_{ni}^2 + \mu K_1M_i^2\omega_{ni}^2 \right] \\
 & + \lambda [2K_1K_{TMD}M_i(-\xi_{ae} + \xi_{ni})\omega_{ni} + C_{TMD}K_{TMD}M_i\omega_{ni}^2] \\
 & + K_1K_{TMD}M_i\omega_{ni}^2 = 0 \\
 & \text{and} \\
 & b\lambda^2 + \lambda C_{TMD} + K_1 \neq 0
 \end{aligned} \tag{A4}$$

APPENDIX B: PERFORMANCE INDEXES

1. IBD1:

$$I_{IBD1} = \frac{\gamma(1+\mu)\xi_{TMD}}{\mu} + \frac{1 - (2+\mu)\gamma^2 + (1+\mu)^2\gamma^4}{4\mu\gamma\xi_{TMD}} + \frac{(\delta^2 + a_1\delta)}{4\mu\gamma\xi_{TMD}}, \tag{B1}$$

where

$$a_1 = -2[(1+\mu)\gamma^2 - 1]. \tag{B2}$$

2. IBD2:

$$I_{IBD2} = \frac{\gamma(1+\mu)\xi_{TMD}}{\mu} + \frac{1 - (2+\mu)\gamma^2 + (1+\mu)^2\gamma^4}{4\mu\gamma\xi_{TMD}} + \left(\frac{a_2}{\delta^2} + \frac{a_1}{\delta} \right) \xi_{TMD}, \tag{B3}$$

where

$$\begin{aligned}
 a_2 &= \frac{\gamma}{\mu} [(1+\mu)^3\gamma^4 - 2(1+\mu)\gamma^2 + 1], \\
 a_1 &= \frac{\gamma}{\mu} [2 + \mu - 2(1+\mu)^2\gamma^2].
 \end{aligned} \tag{B4}$$

3. IBD3:

$$I_{IBD3} = \frac{\gamma(1+\mu)\xi_{TMD}}{\mu} + \frac{1 - (2+\mu)\gamma^2 + (1+\mu)^2\gamma^4}{4\mu\gamma\xi_{TMD}} + \left(\frac{a_2}{\delta^2} + \frac{a_1}{\delta} \right) \xi_{TMD}, \tag{B5}$$

where

$$\begin{aligned} a_2 &= \frac{[1-2\gamma^2+(1+\mu)\gamma^4]}{\mu\gamma^3v^4} - \frac{2[1-(2+\mu)\gamma^2+(1+\mu)^2\gamma^4]}{\mu\gamma v^2} + \frac{\gamma[1-2(1+\mu)\gamma^2+(1+\mu)^3\gamma^4]}{\mu}, \\ a_1 &= \frac{-2[1-(1+\mu)\gamma^2]}{\mu\gamma v^2} - \frac{\gamma}{\mu}[2(1+\mu)^2\gamma^2-2-\mu]. \end{aligned} \quad (\text{B6})$$

4. IBD4:

$$\begin{aligned} I_{\text{IBD4}} &= \frac{\gamma(1+\mu)\xi_{\text{TMD}}}{\mu} + \frac{1-(2+\mu)\gamma^2+(1+\mu)^2\gamma^4}{4\mu\gamma\xi_{\text{TMD}}} + \left(\frac{c_2}{\delta^2} + \frac{c_1}{\delta}\right)\xi_{\text{TMD}} \\ &+ \frac{b_2v^4\delta^2 + b_1\delta + a_2v^4 + a_1v^2}{\xi_{\text{TMD}}}, \end{aligned} \quad (\text{B7})$$

where

$$\begin{aligned} c_2 &= \frac{\gamma}{\mu}[1-(2+\mu)\gamma^2+(1+\mu)^3\gamma^4], \\ c_1 &= \frac{\gamma}{\mu}[(2+\mu)-2(1+\mu)^2\gamma^2], \\ b_2 &= \frac{\gamma^3(1+\mu)^2}{4\mu}, \\ b_1 &= \frac{\gamma^3}{2\mu}v^4[(1+\mu)-(1+\mu)^3\gamma^2] + \frac{\gamma}{4\mu}v^2[2(1+\mu)^2\gamma^2-\mu-2], \\ a_2 &= \frac{\gamma^3}{4\mu}[(1+\mu)^4\gamma^4+(\mu-2)(1+\mu)^2\gamma^2+1], \\ a_1 &= -\frac{\gamma}{2\mu}[(1+\mu)^3\gamma^4-2(1+\mu)\gamma^2+1]. \end{aligned} \quad (\text{B8})$$